



WHEN POSITIVE WORDS MEAN NEGATIVE: IMPROVING ROBUSTNESS TO SARCASM IN TWITTER SENTIMENT ANALYSIS

A COMPARATIVE STUDY OF MULTI-TASK AND
RATIONALE SUPERVISION USING ROBERTA

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Abstract

Strong benchmark performance can hide systematic weaknesses on linguistically difficult inputs. In Twitter sentiment analysis, sarcasm is a major challenge because the intended meaning often differs from the literal wording. Our preliminary experiment shows that a state-of-the-art RoBERTa-based Twitter sentiment model performs strongly on the TweetEval benchmark (macro-F1 = 0.719), but drops to 0.557 on a sarcasm-heavy out-of-distribution evaluation set, revealing a clear robustness failure on sarcastic language.

Prior research has mainly addressed sarcasm-aware sentiment analysis through Multi-Task Learning (MTL) with sarcasm detection, often using earlier architectures and only standard in-distribution evaluations. Meanwhile, Rationale Supervision (RS), which guides models using natural language explanations, has shown promise for sentiment classification but remains underexplored for sarcasm-aware Twitter sentiment analysis. Therefore, this thesis investigates whether these auxiliary signals improve robustness to sarcasm by comparing four RoBERTa-based variants: a sentiment-only Baseline, an MTL model with sarcasm detection, an RS model using LLM-generated rationales distilled into the encoder, and a Combined model integrating both approaches.

Results show that RS provides the most reliable robustness improvement, increasing macro-F1 on the sarcasm-heavy set from 0.557 to 0.701, a 26% relative improvement over the Baseline ($p = 0.008$), while maintaining strong benchmark performance. The Combined model achieves the highest out-of-distribution performance (0.724 macro-F1) but with higher seed variance, while MTL provides only modest and statistically unreliable gains. Overall, the findings suggest that reasoning-based supervision improves sarcasm robustness more effectively than sarcasm detection alone, with implications for more reliable sentiment analysis systems.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

The datasets used in this thesis were obtained from publicly available academic sources released for research purposes. The TweetEval sentiment dataset (Barbieri et al., 2020; Rosenthal et al., 2017) was accessed through the [TweetEval GitHub repository](#). The iSarcasm dataset (Oprea & Magdy, 2020) was obtained from its [official repository](#). The sarcasm-heavy out-of-distribution evaluation set was constructed from the Twitter sarcasm subset of SemEval-2014 Task 9 (Rosenthal et al., 2014), obtained from the [SemEval website](#). No new data was collected from human or animal participants.

All datasets consist of social media text released for academic research. Since the data originates from Twitter/X, it may contain demographic, linguistic, and cultural biases and should not be interpreted as representative of the broader population. During preprocessing, usernames were replaced with a generic @user token where needed to further reduce identifiable information.

All figures and visualizations were created by the author using Python visualization libraries and draw.io. The implementation code is publicly available in the [project GitHub repository](#).

The project was implemented in Python using PyTorch, Hugging Face Transformers, scikit-learn, NumPy, Pandas, Matplotlib, Optuna, SentenceTransformers, and related libraries. Library version numbers are listed in Appendix A. The models build on RoBERTa (Liu et al., 2019) using the [cardiffnlp/twitter-roberta-base-sentiment-latest](#) checkpoint through Hugging Face Transformers (Wolf et al., 2020). Rationale generation was performed using the OpenAI Batch API with gpt-4o-mini, while rationale supervision used sentence embeddings from all-MiniLM-L6-v2 via SentenceTransformers (Reimers & Gurevych, 2019).

Generative AI tools were used only as supportive assistants during the project. ChatGPT was used for grammar checking and clarity improvements of the author’s original writing, while Gemini was used for debugging support and code optimization suggestions. All outputs were reviewed and verified by the author. The thesis was written in LaTeX using Overleaf, references were managed with BibTeX, and model training and tuning were conducted on Tilburg University’s GPU infrastructure.

2 INTRODUCTION

2.1 Problem Statement

Sentiment analysis is one of the major tasks in Natural Language Processing (NLP), focusing on classifying the emotional tone of text as positive,

neutral, or negative (Liu, 2012). The task is especially important in social media settings such as Twitter (now known as X), where sentiment signals are widely used for applications including brand monitoring and public opinion analysis. However, social media language is often informal, noisy, and highly contextual, making sentiment analysis more challenging than in formal writing domains (Wankhade et al., 2022).

Among the linguistic phenomena that challenge sentiment analysis, sarcasm remains particularly difficult because the literal wording often differs from the intended meaning (Maynard & Greenwood, 2014). For example, the tweet *"Great, my flight is delayed again"* contains positive lexical cues but actually expresses negative sentiment. Correctly interpreting sarcasm therefore requires contextual reasoning and speaker intent beyond surface words alone (Joshi et al., 2017). This makes sarcasm an important test case for whether sentiment models rely on genuine contextual understanding or superficial lexical patterns (Geirhos et al., 2020; McCoy et al., 2019).

Transformer models such as RoBERTa (Liu et al., 2019) achieve strong performance on Twitter sentiment benchmarks such as TweetEval (Barbieri et al., 2020). However, strong benchmark performance can conceal systematic weaknesses on linguistically challenging inputs. Initial experiments of this thesis reveal a substantial generalization gap: a strong RoBERTa-based Twitter sentiment model achieves 0.719 macro-F1 on the standard TweetEval test set, but drops to 0.557 on a sarcasm-heavy out-of-distribution set, a decrease of more than 16 macro-F1 points. This suggests that sarcasm remains a systematic failure mode rather than random noise, motivating a closer investigation into methods for improving robustness to sarcastic language.

One popular direction is **Multi-Task Learning (MTL)**, where sentiment classification is jointly trained with sarcasm detection to encourage shared representations between the two tasks (Caruana, 1997; Majumder et al., 2019). Prior work reports improved sentiment performance from this setup (Majumder et al., 2019; Tan et al., 2023), but mainly with earlier architectures such as Bi-LSTM and GRU under standard in-distribution evaluation. The second direction explored in this study is **Rationale Supervision (RS)**, where LLM-generated rationales are used as auxiliary training signals to encourage the model to align its internal representation with sentiment-relevant reasoning. This approach is inspired by rationale-based distillation, where reasoning generated by large language models (LLMs) is used as additional supervision to train smaller models without requiring the LLM at inference time (Hsieh et al., 2023). Recent work also suggests that rationale-based supervision can improve sentiment classification (Nguyen et al., 2025), yet its effectiveness for sarcasm robustness remains underexplored. As a result, it remains unclear whether these

strategies improve robustness in modern Twitter sentiment transformers under sarcasm-heavy out-of-distribution conditions.

Both approaches are forms of *auxiliary supervision*, where the main sentiment task is trained together with additional signals intended to improve robustness to sarcasm. However, they target the problem from different angles: MTL uses sarcasm labels to teach the model when sarcasm is present, while RS uses textual explanations to guide how sentiment should be interpreted. These approaches have not been directly compared within a unified transformer framework, leaving it unclear whether their effects are complementary when combined.

2.2 Research Strategy

Motivated by these gaps, this thesis investigates whether auxiliary supervision can improve the robustness of a RoBERTa-based sentiment classifier on sarcastic tweets. Three strategies are compared against a sentiment-only baseline: Multi-Task Learning (MTL) with sarcasm detection, Rationale Supervision (RS) using LLM-generated explanations, and a Combined model integrating both signals.

The strategy is designed to evaluate robustness under distribution shift. All models are trained on standard TweetEval sentiment data, then evaluated on both the in-distribution TweetEval test set and a sarcasm-heavy out-of-distribution set. Exploratory data analysis motivates key preprocessing and modeling decisions, while pilot experiments compare alternative strategies such as oversampling and data augmentation before full-scale training.

Beyond aggregate scores, the study analyzes error patterns and statistical reliability to examine not only whether auxiliary supervision improves robustness, but also how it changes model behavior on sarcastic inputs.

To address these objectives, the following main research question is proposed:

To what extent can auxiliary supervision through sarcasm detection, rationale supervision, and their combination improve the out-of-distribution robustness of a RoBERTa-based sentiment model on sarcastic tweets?

This is broken down into five sub-questions:

RQ1 *How does a baseline RoBERTa model perform on standard (in-distribution) and sarcasm-heavy (out-of-distribution) test sets, and what error patterns emerge on sarcastic tweets?*

RQ2 *To what extent does Multi-Task Learning with sarcasm detection improve sentiment classification performance on sarcasm-heavy data relative to the baseline?*

RQ3 *To what extent does Rationale Supervision improve sarcasm-heavy out-of-distribution performance compared to the baseline and Multi-Task Learning?*

RQ4 *Does combining sarcasm detection and rationale supervision lead to further improvements beyond either approach individually?*

RQ5 *To what extent are the observed robustness differences statistically reliable given the limited size of the sarcasm-heavy out-of-distribution set ($n = 86$)?*

RQ1 establishes the baseline failure mode and error patterns, RQ2 and RQ3 evaluate each auxiliary supervision strategy independently, RQ4 evaluates their combination, and RQ5 assesses the statistical reliability of the results given the limited size of the out-of-distribution evaluation set. Together, these sub-questions provide a structured empirical strategy for answering the main research question.

2.3 Societal Relevance

Sentiment analysis systems are increasingly used in areas such as customer feedback analysis, political opinion tracking, and crisis monitoring. When sarcastic complaints or criticism are misclassified as positive sentiment, the resulting analyses can produce misleading conclusions about customer satisfaction or public opinion. For example, political sarcasm such as “Great job, another tax increase” may be incorrectly interpreted as positive sentiment, causing systems to underestimate public criticism. Because sarcasm can be expressed differently across cultural, linguistic, and political contexts (Dress et al., 2008), sentiment models may interpret some groups’ opinions less accurately than others. As a result, certain forms of criticism or dissatisfaction may be systematically underestimated. Improving robustness to sarcasm is therefore important for building more reliable and fair sentiment analysis systems, particularly in political communication and crisis analysis.

2.4 Scientific Relevance

Scientifically, this study contributes to research on robust sentiment analysis under linguistic distribution shift. It provides one of the first controlled comparisons between Multi-Task Learning with sarcasm detection and reasoning-based supervision using LLM-generated rationales in a modern Twitter-adapted transformer framework. More broadly, the study frames

sarcasm robustness not only as a data problem, but also as a supervision problem: different auxiliary signals may lead models to handle sarcastic language in different ways. Beyond benchmark performance, the thesis also analyzes prediction errors and internal embedding behavior to better understand why some supervision strategies improve robustness more effectively than others.

2.5 *Main Findings*

The results show that auxiliary supervision improves robustness to sarcastic tweets, but its effectiveness varies by approach. Rationale Supervision (RS) provides the most reliable improvement, increasing sarcasm-heavy macro-F1 from 0.557 to 0.701 while preserving standard TweetEval performance. The Combined model achieves the highest out-of-distribution performance (0.724 macro-F1), but with weaker stability across seeds, while Multi-Task Learning (MTL) provides only modest gains. Overall, the findings suggest that reasoning-based supervision is more effective than sarcasm detection alone for robust sentiment analysis.

3 RELATED WORK

3.1 *Transformer Models for Twitter Sentiment Analysis*

Sentiment analysis on Twitter presents unique challenges compared to formal-domain text due to the informal, noisy, and rapidly evolving nature of tweets (Wankhade et al., 2022).

The introduction of the Transformer architecture (Vaswani et al., 2017) significantly advanced NLP through self-attention mechanisms that capture long-range contextual dependencies. Building on this, BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019) established the pre-train–fine-tune paradigm and achieved state-of-the-art performance across many NLP tasks, including sentiment analysis. However, these models are pretrained mainly on general-domain corpora such as Wikipedia and BookCorpus, which differ substantially from Twitter language (Nguyen et al., 2020).

To address this domain mismatch, Twitter-specific language models have been developed. BERTweet (Nguyen et al., 2020), trained on 850 million English tweets, improves performance across Twitter NLP tasks. TimeLMs (Loureiro et al., 2022) further extend this approach through temporally adaptive pretraining, showing that incorporating more recent training data helps capture the evolving nature of social media language. The most recent model in the TimeLMs family is `cardiffnlp/twitter-`

roberta-base-sentiment-latest¹. The model is pretrained on 124 million tweets from 2018 to 2021 and further fine-tuned on TweetEval sentiment (Barbieri et al., 2020). He (2024) reports that this model achieves an F1 score of 0.715, slightly outperforming BERTweet (0.706), motivating its use as the baseline model in this study.

Despite this progress, existing evaluations remain focused on aggregate benchmark performance, which may hide important model weaknesses under linguistically challenging conditions (McCoy et al., 2019; Ribeiro et al., 2020). This motivates a closer examination of sarcasm as a systematic failure mode in sentiment analysis.

3.2 Sarcasm as a Systematic Failure Mode

Sarcasm remains a persistent challenge for sentiment analysis because the literal wording often differs from the intended meaning. Sarcastic tweets frequently contain lexical cues that conflict with the actual sentiment, causing models to make confident but incorrect predictions. Because sarcasm often depends on contextual and pragmatic interpretation rather than literal wording (Joshi et al., 2017), it provides a direct test of whether sentiment models capture intended meaning or mainly rely on surface-level sentiment cues.

This difficulty persists across sentiment analysis models. Farha and Magdy (2020) show that state-of-the-art Arabic sentiment models perform substantially worse on sarcastic tweets than non-sarcastic ones, suggesting that sarcasm is a consistent weakness rather than random noise. Similarly, Bhargava et al. (2025) show that filtering sarcastic content before classification can improve accuracy on sarcasm-heavy data by up to 21%. However, the challenge is not how to remove sarcastic content, but how to train models that handle it correctly.

Some prior work addresses sarcasm through context augmentation, such as incorporating dialogue history and speaker information (Helal et al., 2024), or hybrid architectures combining transformers with graph neural networks (Qin et al., 2025). However, these approaches often require additional metadata or more complex architectures rather than improving robustness through standard fine-tuning. This motivates exploring supervision strategies that improve sarcasm robustness directly through the training objective.

¹ <https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest>

3.3 *Multi-Task Learning for Sarcasm-Aware Sentiment Analysis*

One supervision-based approach to sarcasm robustness is Multi-Task Learning (MTL). Introduced by Caruana (1997), MTL trains a shared encoder on related tasks simultaneously so that useful information from one task can support another. In sarcasm-aware sentiment analysis, the intuition is that sarcasm detection can help models rely less on literal sentiment words and better recognize when intended meaning differs from the wording.

Prior work supports this intuition. Majumder et al. (2019) report 3–4% F1 improvements using a joint sarcasm–sentiment GRU model on Twitter data. Tan et al. (2023) similarly report gains from using sarcasm detection as an auxiliary task in a Bi-LSTM framework, while El Mahdaouy et al. (2021) extend MTL to a BERT-based Arabic Twitter setting. However, most existing evidence comes from earlier architectures, different language settings, or standard in-distribution datasets. It therefore remains unclear whether MTL improves robustness to sarcasm in modern Twitter sentiment transformers on sarcasm-heavy out-of-distribution data.

A further limitation of MTL is the risk of negative transfer, where auxiliary supervision decreases performance on the primary task (Crawshaw, 2020). This risk is especially relevant when the sarcasm dataset is much smaller than the sentiment dataset, as in the work of Tan et al. (2023), where the sarcasm dataset is roughly 10 times smaller. Yet prior work does not explicitly evaluate on sarcasm-heavy out-of-distribution data or address the large imbalance between sarcasm and sentiment training data.

3.4 *Rationale Supervision via Distillation from LLM-Generated Rationales*

While MTL improves robustness through auxiliary sarcasm-task supervision, rationale supervision instead guides the model using natural language explanations for why a sentiment label is correct (Zaidan et al., 2007).

Prior work shows that rationale models can become more robust to irrelevant or adversarial text on product review data by filtering out distracting parts of the input (Chen et al., 2022), suggesting that rationale supervision can reduce reliance on misleading surface-level sentiment cues. This is especially relevant for sarcasm, where literal wording often conflicts with intended sentiment. Similarly, Nguyen et al. (2025) show that human-written rationales improve healthcare sentiment classification by 2% macro-F1. However, these studies are conducted in relatively formal domains where sentiment is more explicit than in sarcasm-heavy Twitter data, and human-written rationales are costly to collect at scale.

Large language models offer a scalable alternative by automatically generating explanations. Hsieh et al. (2023) show that LLM-generated

rationales can be used as additional supervision to train smaller models without requiring the LLM at inference time. However, their work focuses on training models to generate rationales. In contrast, this thesis investigates whether rationale information can improve RoBERTa, an encoder-based sentiment classifier, without converting it into a generative model. This keeps the focus on sentiment prediction while still using rationales as auxiliary supervision.

Rationales are also not automatically useful. Their effectiveness depends on rationale quality and whether they provide information beyond the label itself (Wiegrefe et al., 2021). This is especially important for LLM-generated rationales, which may be hallucinated, inconsistent, or overly generic (Camburu et al., 2018; Ji et al., 2023). It therefore remains unclear whether LLM-generated rationales can provide reliable supervision for improving sarcasm robustness in Twitter sentiment analysis.

3.5 Research Gap

In summary, three main gaps emerge from the current literature.

First, existing MTL studies for sarcasm-aware sentiment analysis focus mainly on earlier architectures or standard in-distribution evaluation, making it unclear whether their gains transfer to modern Twitter sentiment transformers under distribution shift.

Second, rationale supervision has shown promise in sentiment analysis, but its effectiveness for improving robustness to sarcastic language remains largely unexplored.

Third, although MTL and rationale supervision address the same robustness problem through different supervision signals, to our knowledge, no prior work has examined their combination within a unified transformer framework. Therefore, little is known about whether their effects are complementary when combined.

This thesis addresses these gaps through a controlled comparison of MTL, rationale supervision, and their combination under sarcasm-heavy out-of-distribution evaluation, while also examining how different supervision signals affect robustness, prediction errors, and model behavior on sarcastic inputs.

4 METHODS

This section translates the research questions into a controlled experimental design. As shown in Figure 1, the pipeline consists of preprocessing, rationale generation with GPT-4o-mini, model architecture design, model

training across three random seeds, and evaluation on both TweetEval and the sarcasm-heavy out-of-distribution set.

We train on standard sentiment data and evaluate on sarcasm-heavy out-of-distribution data because this provides a stronger test of robustness than training directly on sarcastic tweets. In the latter setting, performance gains may partly reflect learning dataset-specific sarcasm patterns rather than genuine generalization.

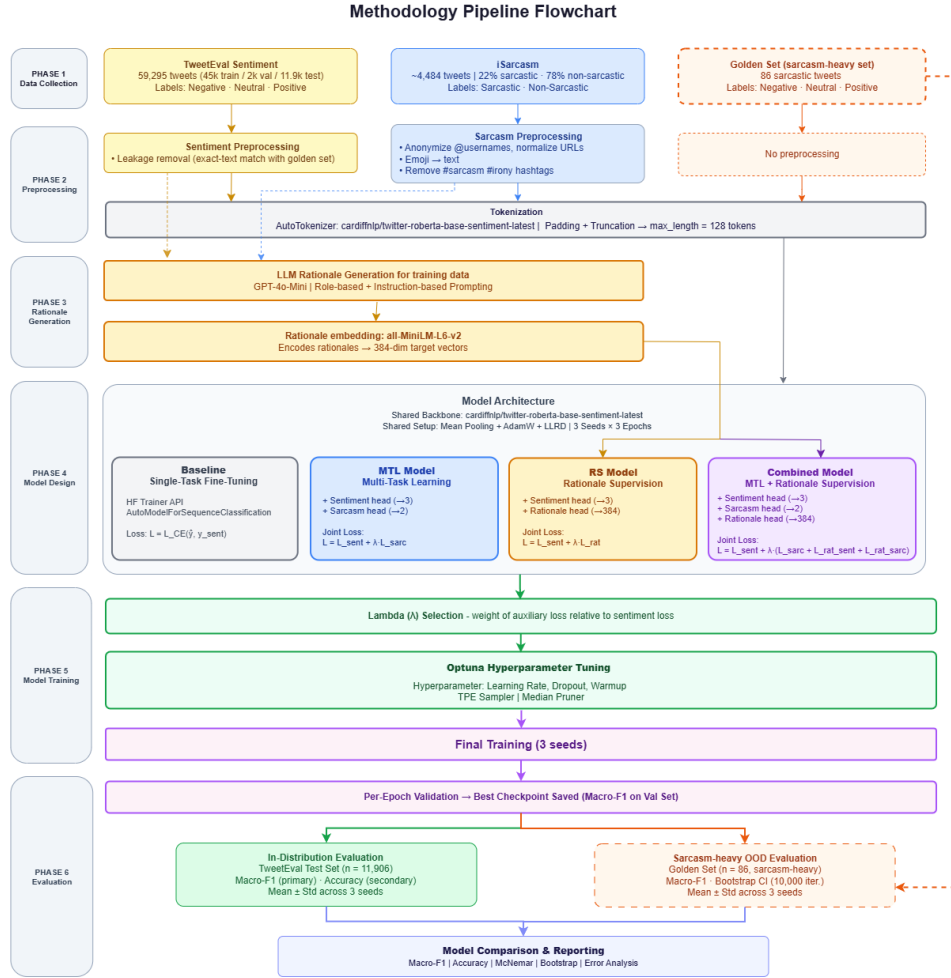


Figure 1: Research Methodology Flowchart

4.1 Dataset Description

This study uses three datasets, each serving a different role in training and evaluation:

(1) **TWEETEVAL SENTIMENT.** TweetEval Sentiment (Barbieri et al., 2020), derived from SemEval-2017 Task 4 (Rosenthal et al., 2017), is used as the primary dataset for training and in-domain evaluation. It consists of English tweets labeled as Negative (0), Neutral (1), or Positive (2), and split by original author into 45,615 training, 2,000 validation, and 12,284 test samples. The dataset is moderately imbalanced, with Neutral as the majority class. TweetEval is selected because it is a widely used Twitter sentiment benchmark and aligns directly with the pretrained backbone used in this study.

(2) **ISARCASM.** The auxiliary dataset is iSarcasm (Oprea & Magdy, 2020), used for sarcasm detection in the multi-task setting. It contains tweets collected between 2008 and 2019 through a web platform where users self-annotated their own posts as sarcastic or non-sarcastic. This reduces annotation noise compared to distant supervision methods that automatically label sarcasm using hashtags alone. It contains approximately 4,850 tweets, with about 22% labeled sarcastic and 78% non-sarcastic, reflecting the naturally imbalanced distribution of sarcasm in social media data.

(3) **GOLDEN EVALUATION SET.** The out-of-distribution evaluation set consists of 86 sarcastic tweets with three-class sentiment labels from SemEval-2014 Task 9 Subtask B (Rosenthal et al., 2014). The tweets were originally collected using sarcasm-related hashtags and are used here as a sarcasm-heavy robustness benchmark, referred to throughout this thesis as the Golden Set.

4.2 *Preprocessing*

Different preprocessing strategies are applied to each dataset due to differences in normalization and their roles in the pipeline.

4.2.1 *Sentiment Data*

The TweetEval dataset is already normalized by the original authors, with usernames anonymized and URLs removed. Because the tokenizer used in this thesis is already pretrained on Twitter text, additional normalization such as lowercasing or punctuation removal could remove useful sentiment cues. Therefore, no further preprocessing is applied besides overlap removal with the Golden Set.

4.2.2 Data Leakage Removal

Because both TweetEval and the Golden Set originate from Twitter, there is a risk that some tweets may appear in both datasets. During exploratory data analysis, we identified 57 overlapping tweets. This could cause the model to memorize evaluation samples rather than generalize to unseen sarcastic tweets, constituting a form of data leakage (Kaufman et al., 2012). Therefore, all overlapping tweets are removed from the TweetEval training set.

4.2.3 Sarcasm Data

Unlike TweetEval, the iSarcasm dataset is not pre-normalized and therefore requires additional preprocessing. Sarcasm-related hashtags such as #sarcasm, #sarcastic, and #irony are removed to prevent the model from relying on trivial keyword cues instead of learning sarcasm patterns. Usernames are anonymized to @user, URLs are standardized to http, and emojis are converted into textual descriptions using the emoji library (Felbo et al., 2017; Khan et al., 2025). Because sarcasm detection is used only as an auxiliary training task rather than a standalone evaluation task, the original train and test splits are merged to maximize the available auxiliary data.

4.2.4 Tokenization

All text is tokenized using the AutoTokenizer from cardiffnlp/twitter-roberta-base-sentiment-latest, with padding and truncation to a maximum length of 128 tokens. This length is sufficient for tweets (Barbieri et al., 2020). The same tokenizer is used across all models to ensure that performance differences come from the auxiliary supervision methods rather than differences in tokenization.

4.3 Rationale Generation

For the RS and Combined models, rationales are generated using GPT-4o-mini (Hurst et al., 2024) through the OpenAI Batch API with temperature set to 0.1 to reduce output randomness. GPT-4o-mini is selected because it provides a balance between reasoning quality and computational cost, making large-scale rationale generation feasible. Rationales are generated once and reused across all experiments.

The prompts combine role-based prompting and constraint prompting (White et al., 2023). Each prompt begins with the role description “*You are an expert sentiment annotator for social media*”, followed by constraints on

rationale content and format. The ground-truth sentiment label is included in the prompt so that the model explains the annotated label rather than predicting it.

Because prior work shows that low-quality rationales may introduce shortcut patterns or generic explanations (Camburu et al., 2018; Ji et al., 2023; Wiegrefe et al., 2021), the prompts are constrained to produce concise, evidence-based explanations.

Five constraints are enforced:

- **Lexical grounding:** Rationales must quote specific words from the tweet to provide token-level evidence.
- **Sarcasm handling:** Rationales must explain mismatches between literal wording and intended meaning.
- **Direct start:** Rationales avoid generic openings like “*This tweet is...*”.
- **Length under 20 words:** Shorter rationales keep the supervision signal focused.
- **Anti-leakage:** Rationales should not mention sentiment labels to prevent trivial label cues.

4.4 Exploratory Data Analysis (EDA)

EDA is conducted to inspect dataset characteristics, motivate methodological choices, and evaluate the quality of the generated rationales. The main findings are summarized below, while detailed results are provided in Appendix C.

First, class distribution analysis shows that TweetEval is moderately imbalanced, with Neutral as the majority class (48.3%). In addition, the Golden Set contains substantially more negative tweets than the standard test set (46.5% vs. 32.3%). This motivates the use of Macro-F1 as the primary evaluation metric and suggests that the Golden Set represents a distribution shift from the standard test set.

To examine the surface-level ambiguity introduced by sarcasm, lexical sentiment polarity is measured using TextBlob (Loria, 2024), a lexicon-based tool that assigns sentiment scores based on words without considering context. As shown in Figure 2, sarcastic tweets show a heavier positive-polarity tail than non-sarcastic tweets. This indicates that sarcastic tweets often contain positive lexical cues even when the intended meaning may differ from the literal wording, highlighting the challenge sarcasm poses for sentiment classification.

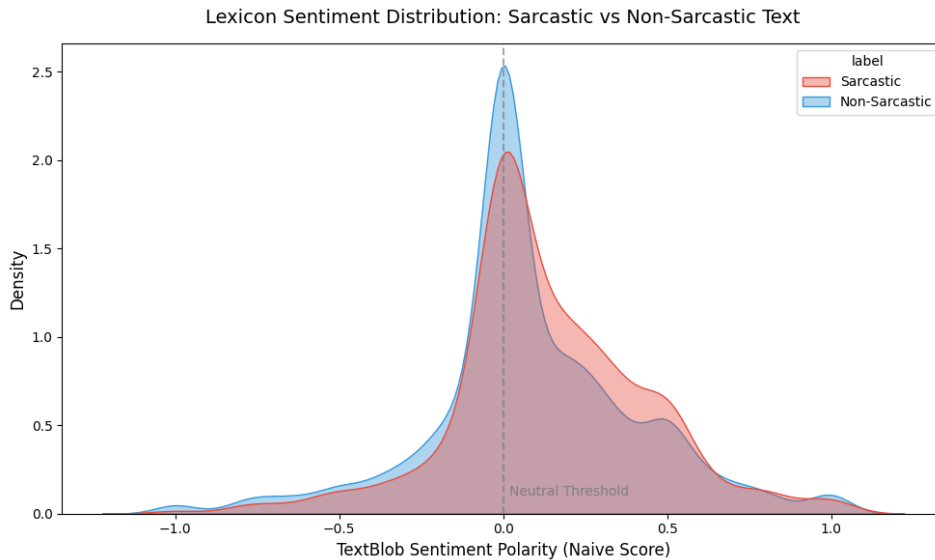


Figure 2: Distribution of TextBlob sentiment polarity scores for sarcastic and non-sarcastic tweets. Scores range from -1 (negative) to $+1$ (positive). Sarcastic tweets show a heavier positive-polarity tail, indicating that many sarcastic tweets contain positive lexical cues.

Vocabulary overlap analysis using the top-1000 most frequent tokens from each dataset shows that 54% of tokens are shared between the sentiment and sarcasm datasets. This indicates partially overlapping lexical spaces between the two tasks, supporting the use of a shared encoder in the multi-task setting.

Finally, rationale quality is evaluated using semantic similarity with all-MiniLM-L6-v2 embeddings. The mean cosine similarity between tweets and generated rationales is 0.45, indicating that the rationales remain grounded in the source text while still providing additional information rather than simply repeating the input.

4.5 Model Architectures

This study follows a controlled ablation-style design in which all architectural and optimization settings remain fixed across variants, while only the supervision signal changes. All four models use the pretrained backbone `cardiffnlp/twitter-roberta-base-sentiment-latest`² (Loureiro et al., 2022), chosen because it is pretrained on Twitter data and further fine-tuned on the TweetEval benchmark, providing a strong domain-consistent initialization for sentiment classification.

² <https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest>

Alternative backbones such as BERTweet are intentionally excluded because changing the tokenizer, pretraining data, or model capacity would introduce additional confounding factors. Keeping the backbone fixed makes it easier to isolate the effect of auxiliary supervision. Section 4.4 further supports the use of a shared encoder across sentiment and sarcasm supervision.

Within this controlled setup, all variants share three key design choices:

(1) **MEAN POOLING.** The standard RoBERTa model uses the CLS token, a special token at the beginning of the sequence intended to represent the entire input (Liu et al., 2019). However, sarcastic cues may appear throughout the tweet rather than at a single position. Therefore, mean pooling is used instead, which averages token-level representations across the sequence. The final representation is computed as:

$$\mathbf{h} = \frac{\sum_{i=1}^L \mathbf{h}_i \cdot m_i}{\sum_{i=1}^L m_i} \quad (1)$$

where $\mathbf{h}_i$ is the hidden state of token i , m_i the attention mask, and L the sequence length. Recent NLP work also suggests that mean pooling can outperform single-token pooling for classification tasks (Ennadir et al., 2025).

(2) **LAYER-WISE LEARNING RATE DECAY (LLRD).** Different Transformer layers capture different levels of information: lower layers learn more general linguistic patterns, while upper layers become more task-specific during fine-tuning (Jawahar et al., 2019). Fine-tuning all layers with the same learning rate can overwrite useful pretrained knowledge in lower layers (Sun et al., 2020). LLRD applies smaller learning rates to lower layers and larger learning rates to upper layers, helping preserve pretrained knowledge while adapting higher layers to the task. The multipliers are fixed as:

$$\begin{aligned} \text{Embeddings} &: 0.1 \times \eta \\ \text{Layers 1-6} &: 0.2 \times \eta \\ \text{Layers 7-10} &: 0.5 \times \eta \\ \text{Layers 11-12} &: 1.0 \times \eta \\ \text{Task Heads} &: 2.0 \times \eta \end{aligned}$$

These ratios are fixed rather than tuned to keep the optimization setup consistent across model variants and avoid adding another tuning dimension unrelated to the supervision comparison.

(3) **OPTIMIZATION AND REGULARIZATION.** All models use AdamW (Loshchilov & Hutter, 2019) with weight decay (0.01), linear warmup scheduling, gradient clipping (max norm = 1.0), and label smoothing (0.1) (Müller et al., 2020). These techniques are applied to stabilize training and reduce overfitting.

4.5.1 Baseline Model (RoBERTa)

The Baseline fine-tunes `cardiffnlp/twitter-roberta-base-sentiment-latest` under a standard single-task setup, serving as the reference model. The mean-pooled representation is passed directly to a classification head optimized with cross-entropy loss:

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{CE}}(\hat{\mathbf{y}}, y_{\text{sent}}) \quad (2)$$

where $\hat{\mathbf{y}}$ is the predicted sentiment distribution and y_{sent} is the gold sentiment label. Training uses a learning rate of 2×10^{-5} , batch size 16, and 3 epochs.

4.5.2 Multi-Task Learning (MTL) Model

The MTL model extends the Baseline by jointly learning sentiment classification as the primary task and sarcasm detection as an auxiliary task using a shared RoBERTa encoder. On top of the shared encoder, two task-specific heads are added: a sentiment head ($\text{Linear}_{768 \rightarrow 384} \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow \text{Linear}_{384 \rightarrow 3}$) and a sarcasm head with the same structure but output dimension 2. The architecture is illustrated in Figure 3.

By jointly predicting sarcasm, the model is exposed to cases where wording and intended sentiment diverge, encouraging the shared encoder to rely more on contextual meaning.

To balance optimization between tasks, alternating batch updates are used: the model processes sentiment and sarcasm batches separately instead of mixing both tasks within the same batch (Crawshaw, 2020). This helps reduce negative transfer caused by the roughly 10:1 imbalance between the sentiment and sarcasm data (45k vs. 4.5k samples). Over-sampling and data augmentation were explored in pilot experiments (Appendix 12) but did not improve performance, so the final setup repeatedly cycles through the smaller sarcasm dataset during training.

The overall loss is defined as:

$$\mathcal{L}_{\text{MTL}} = \mathcal{L}_{\text{sent}} + \lambda \mathcal{L}_{\text{sarc}} \quad (3)$$

where both terms are cross-entropy losses with label smoothing 0.1, and λ controls the contribution of the auxiliary sarcasm objective. $\lambda = 0.2$ is

selected via the robustness-oriented composite criterion. At inference time, only the sentiment head is used, so any improvement reflects stronger learned representations rather than reliance on auxiliary predictions.

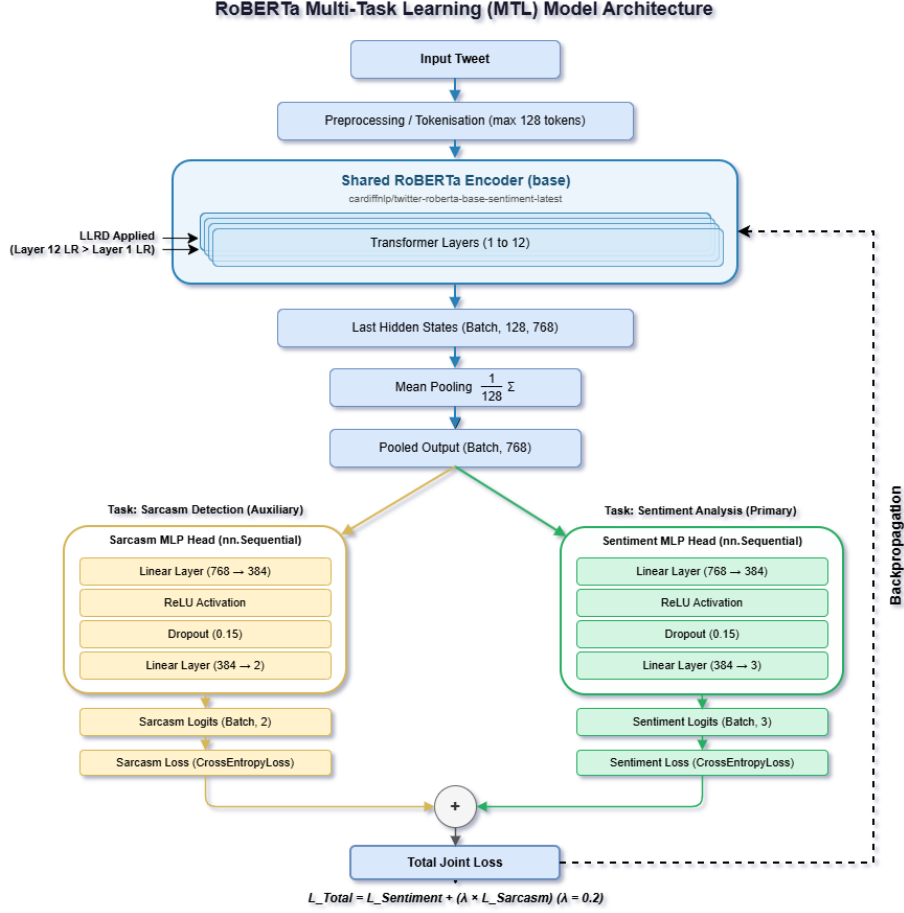


Figure 3: MTL Model Architecture

4.5.3 Rationale Supervision (RS) Model

Instead of learning from sarcasm labels, the RS model learns from LLM-generated explanations associated with each tweet. The encoder is trained to align tweet representations with embeddings of these rationales, encouraging the model to capture the reasoning behind the sentiment label. This follows feature-based knowledge distillation (Hinton et al., 2015), where supervision is provided through continuous embeddings rather than discrete labels.

The architecture extends the shared RoBERTa encoder with two heads: a sentiment classification head, identical to the MTL sentiment head, and a rationale head implemented as a two-layer MLP (Linear_{768→384} → ReLU →

Dropout $\rightarrow$ Linear_{384 $\rightarrow$ 384}). The output dimension is set to 384 to match the embedding size of the frozen teacher model. The architecture is illustrated in Figure 4.

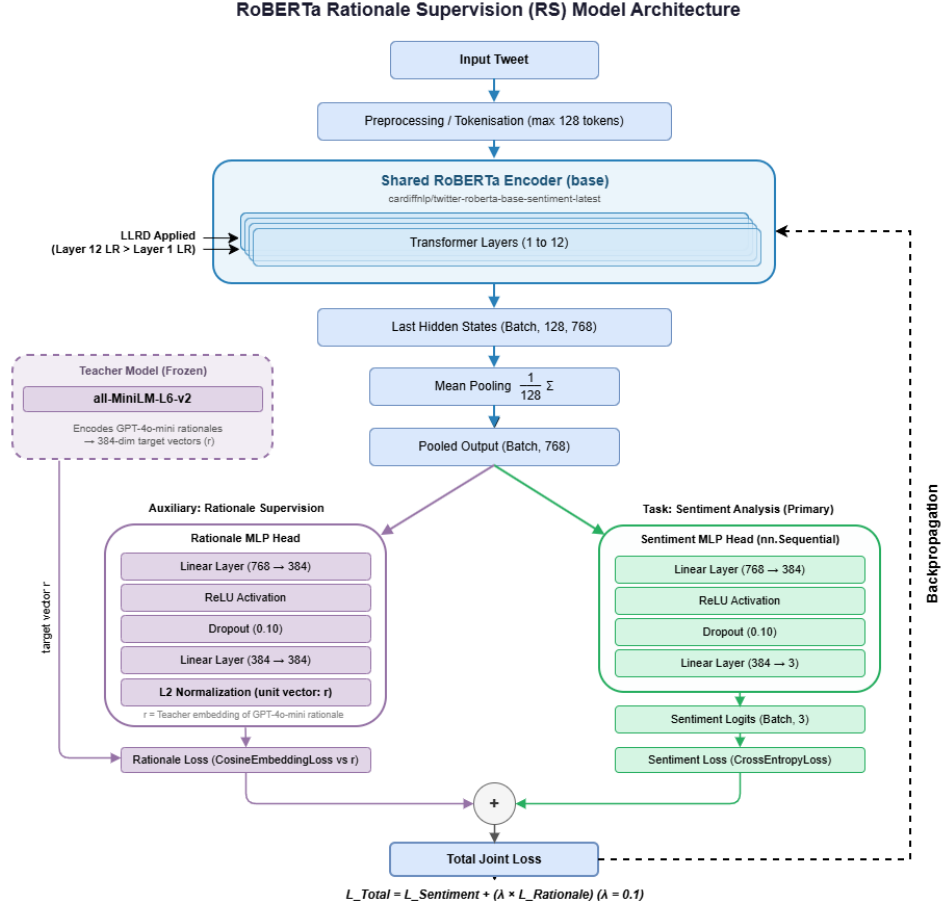


Figure 4: RS Model Architecture

To provide these target rationale embeddings, a separate pretrained sentence encoder is used as a teacher model. The teacher model is all-MiniLM-L6-v2 (Wang et al., 2021), a lightweight Sentence Transformer selected for its balance between performance and computational efficiency, making it practical for encoding approximately 50,000 rationales without the computational cost of larger models such as all-mpnet-base-v2 (Song et al., 2020).

During training, the frozen teacher model encodes the rationale, while the student sentiment encoder processes the tweet. The model is then trained to align these two representations, encouraging tweet embeddings to reflect rationale semantics.

Importantly, rationales are used only during training to distill semantic reasoning patterns into the smaller sentiment model. This avoids generating explanations for every new sample during inference, improving efficiency and enabling standalone deployment.

The output of the rationale head is L2-normalized before cosine alignment so that similarity depends on semantic direction rather than embedding scale. The rationale alignment loss is defined as:

$$\mathcal{L}_{\text{rat}} = 1 - \cos(\hat{\mathbf{r}}, \mathbf{r}) \quad (4)$$

where $\hat{\mathbf{r}}$ is the predicted embedding and $\mathbf{r}$ is the frozen teacher embedding of the corresponding rationale.

The final RS objective combines sentiment classification and rationale alignment:

$$\mathcal{L}_{\text{RS}} = \mathcal{L}_{\text{sent}} + \lambda \mathcal{L}_{\text{rat}} \quad (5)$$

The weight $\lambda = 0.1$ is selected for RS, and it is lower than in MTL ($\lambda = 0.2$), reflecting the noisier nature of LLM-generated rationales. A larger λ risks overfitting to stylistic patterns rather than underlying semantics.

4.5.4 Combined Model

The Combined model is the final architectural iteration, integrating both sarcasm classification (MTL) and rationale supervision (RS) into a single framework. A shared RoBERTa encoder is followed by three task-specific heads: a sentiment head (3-class), a sarcasm head (2-class), and a 384-dimensional rationale head. The architecture is illustrated in Figure 5.

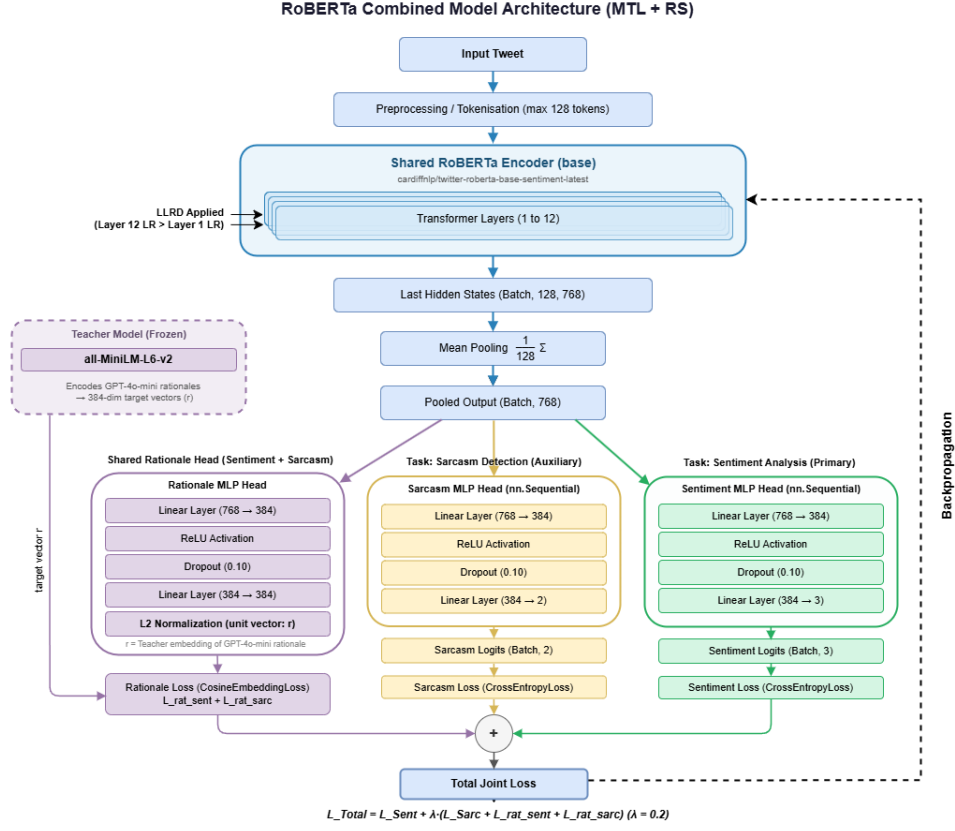


Figure 5: Combined Model Architecture

Training follows the same alternating batch strategy as the MTL model. A single shared rationale head is used for both sentiment and sarcasm rationales, encouraging tweet representations from both tasks to align within the same embedding space. This directly tests whether sentiment and sarcasm supervision share an underlying structure that improves generalization.

For this model, we extend rationale supervision by generating explanations for the sarcasm dataset in addition to the sentiment data. While sentiment rationales explain polarity, sarcasm rationales capture the pragmatic flip where surface wording contradicts intent. The joint objective is defined as:

$$\mathcal{L}_{\text{Combined}} = \mathcal{L}_{\text{sent}} + \lambda \cdot (\mathcal{L}_{\text{sarc}} + \mathcal{L}_{\text{rat_sent}} + \mathcal{L}_{\text{rat_sarc}}) \quad (6)$$

where all auxiliary components share a single weight $\lambda = 0.2$. Due to computational constraints, a shared λ is used, since independently selecting separate auxiliary weights for the Combined model would substantially increase training cost.

The Combined model is the most comprehensive architecture in this study, designed to test whether sarcasm labels and rationales are complementary. If sarcasm supervision is already sufficient, rationale supervision should provide limited additional benefit, and vice versa.

4.6 Experimental Setup

4.6.1 Lambda Selection

In this study, λ controls how strongly the auxiliary objective influences the main sentiment task. The range $\{0.1, 0.15, 0.2, 0.25, 0.3\}$ is selected to allow the auxiliary signal to influence the shared encoder without dominating the main sentiment objective. Since the goal is to improve sarcasm robustness while preserving standard sentiment performance, λ is selected using $(0.4 \times \text{Val F1} + 0.6 \times \text{Gold F1})$, giving slightly greater weight to robustness performance on the sarcasm-heavy Golden Set. This follows robustness evaluation work suggesting that multiple metrics provide a more complete view of model behavior under distribution shift (Calderon et al., 2024). Since the Golden Set contributes to this selection process, its interpretation is discussed as a limitation in Section 6.8.

4.6.2 Hyperparameter Tuning

Optuna is used to tune learning rate, dropout, and warmup ratio over 15 trials using TweetEval validation Macro-F1 as the optimization objective. The search space is:

- Learning rate: $[5 \times 10^{-6}, 4 \times 10^{-5}]$
- Dropout: $[0.10, 0.25]$
- Warmup ratio: $[0.05, 0.20]$

Only the MTL model benefited from Optuna tuning, while the rest performed best under the default settings. Therefore, the final hyperparameters for each model are reported in Table 1.

Table 1: Final hyperparameters used for each model.

Hyperparameter	Baseline	MTL	RS	Combined
Learning rate	2×10^{-5}	8.46×10^{-6}	2×10^{-5}	2×10^{-5}
Dropout	0.10	0.15	0.10	0.10
Warmup ratio	0.10	0.05	0.10	0.10

4.6.3 *Training Procedure*

All models are trained for 3 epochs with batch size 16. During initial prototyping, models were trained for 5 epochs but validation performance consistently peaked within 3, so training is limited to 3 epochs for efficiency. Checkpoints are selected based on best validation Macro-F1. The Baseline uses the HuggingFace Trainer API, while MTL, RS, and Combined use custom PyTorch loops to support joint objectives and alternating training updates.

4.6.4 *Reproducibility*

Results are averaged over three random seeds (7, 42, 100). A global `set_seed()` fixes Python, NumPy, and PyTorch random states. Rationale embeddings are precomputed using a fixed `all-MiniLM-L6-v2` model and reused across runs, ensuring that performance differences come from model design rather than embedding variation.

4.6.5 *Implementation*

Models are implemented in Python 3.10 using PyTorch (v2.5.1) and HuggingFace Transformers. Hyperparameter tuning uses Optuna, and rationale embeddings are generated using Sentence-Transformers. Experiments are run on a single NVIDIA RTX A4000 GPU (16GB VRAM) via the GPU4EDU platform at Tilburg University.

4.6.6 *Exploratory Refinement and Alternative Methods*

Several pilot refinements were evaluated on a reduced setup (20% training data, 2 epochs, 1 seed) before finalizing the main models, including auxiliary data augmentation, oversampling, alternative loss functions, and architectural variants for rationale integration. These experiments help identify promising design choices before full-scale multi-seed evaluation. Some improved the pilot setting but did not replicate under full multi-seed training, so the final models retain the architectures described above. Full details are reported in Appendix I.

4.7 *Evaluation Strategy*

Model performance is evaluated on two test sets. The TweetEval test set (11,906 samples) provides in-distribution evaluation, while the Golden Set (86 samples) serves as a sarcasm-heavy out-of-distribution (OOD) benchmark not used during training. This setup evaluates whether auxiliary

supervision improves robustness beyond in-distribution benchmark performance.

The primary metric is Macro-F1, which is appropriate for the imbalanced TweetEval dataset because it weights all classes equally rather than favoring majority-class performance (Sokolova & Lapalme, 2009). Accuracy is reported as a secondary metric, and results across three seeds are summarized as mean $\pm$ standard deviation to assess training stability.

Pairwise model differences are assessed using McNemar’s test (McNemar, 1947), which is designed for comparing paired classifier predictions on the same evaluation samples (Dietterich, 1998). The test is applied to per-sample correct/incorrect predictions for each seed, and the median p-value across runs is reported to reduce sensitivity to seed-specific outliers.

Given the small size of the Golden Set ($n = 86$), performance estimates from a single evaluation split may be unstable. Therefore, uncertainty is estimated using bootstrap resampling with 10,000 iterations (Efron & Tibshirani, 1994). Each iteration resamples the Golden Set with replacement and recomputes Macro-F1, producing an empirical distribution for estimating performance variability. When multiple pairwise comparisons are performed, p-values are adjusted using the Holm-Bonferroni correction to control the family-wise error rate.

4.8 Error Analysis

Aggregate metrics can hide important differences in model behavior. Therefore, the analysis examines complementary aspects of robustness, including error types, prediction confidence, calibration, interpretability, and internal representations.

ERROR TAXONOMY. To better characterize sarcasm-related failures, errors are grouped into three categories: polarity flips (Negative $\leftrightarrow$ Positive), under-sensitive errors (Negative or Positive predicted as Neutral), and over-sensitive errors (Neutral predicted as Negative or Positive). Polarity flips are particularly important for sarcastic text because sarcastic tweets often contain lexical cues associated with the opposite sentiment (Van Hee et al., 2018). Reduction in polarity flip rate is quantified using Cohen’s h (Cohen, 2013), an effect size measure for differences between proportions.

CONFIDENCE AND CALIBRATION. Prediction confidence on polarity-flip errors is compared across models to examine whether auxiliary supervision reduces overconfident mistakes on sarcastic inputs. Reliability diagrams and Expected Calibration Error (ECE) (Guo et al., 2017) are fur-

ther used to evaluate whether prediction confidence aligns with actual correctness, where lower ECE indicates better calibration.

INTERPRETABILITY VIA INTEGRATED GRADIENTS. To qualitatively examine how auxiliary supervision changes model decision behavior, we apply Integrated Gradients (Sundararajan et al., 2017), an attribution method that estimates each token’s contribution to the final prediction. Three representative Golden Set cases are analyzed: (1) cases where the Baseline fails but auxiliary models succeed, (2) cases where the Baseline and MTL fail but rationale-based models succeed, and (3) cases where all models fail. This allows comparison of whether auxiliary supervision shifts model attention toward contextually meaningful tokens rather than misleading lexical cues.

DIAGNOSTIC AND REPRESENTATION ANALYSES. A key question is whether auxiliary supervision improves sarcasm robustness through meaningful representational change, or simply by acting as a form of regularization (Crawshaw, 2020). To examine this, three additional analyses are performed.

- **Sarcasm Diagnostic.** The auxiliary sarcasm head is evaluated on the fully sarcastic Golden Set, and its predictions are compared with sentiment outcomes. This tests whether improved sarcasm detection is associated with improved sentiment classification.
- **Centered Kernel Alignment (CKA).** CKA (Kornblith et al., 2019) is a representation similarity metric used to compare high-dimensional neural embeddings. Here, it measures the similarity between encoder representations of each auxiliary model and the Baseline on the Golden Set. Lower similarity suggests that auxiliary supervision changes the learned representations rather than merely stabilizing training.
- **t-SNE Visualization.** While CKA provides a global similarity measure, t-Distributed Stochastic Neighbor Embedding (t-SNE) (Van der Maaten & Hinton, 2008), a non-linear dimensionality reduction technique, projects encoder representations into two dimensions for visualization. The analysis focuses on sarcastic-negative and genuinely positive tweets because both can contain similarly positive wording despite expressing opposite sentiment. These cases represent a common lexical trap for sentiment models. Clearer separation suggests greater reliance on pragmatic meaning rather than surface-level sentiment words.

Together, these analyses provide insight into how auxiliary supervision affects prediction behavior and internal representations under distribution shift.

5 RESULTS

This section presents the performance of the four RoBERTa-based models across the in-distribution **test set** and the out-of-distribution **golden set**. Findings are presented to directly address the research questions regarding baseline failure (RQ₁), the impact of auxiliary signals (RQ₂₋₄), and the statistical reliability of these improvements (RQ₅).

5.1 Main Performance Results

Table 2 reports the mean Macro-F1 and accuracy with standard deviation across three random seeds, with Figure 6 visualizing these results with ± 1 SD error bars.

Table 2: Mean Macro-F1 and accuracy ($\pm$ SD) across three seeds.

Model	Test F1	Test Acc	Gold F1	Gold Acc
Baseline	0.7185 ± 0.0025	0.7174 ± 0.0027	0.5565 ± 0.0391	0.5814 ± 0.0308
MTL	0.7158 ± 0.0030	0.7143 ± 0.0033	0.6429 ± 0.0213	0.6473 ± 0.0135
RS	0.7186 ± 0.0069	0.7193 ± 0.0082	0.7010 ± 0.0092	0.7054 ± 0.0067
Combined	0.7137 ± 0.0033	0.7127 ± 0.0035	0.7244 ± 0.0429	0.7248 ± 0.0408

Note: Best result per column is **bold**.

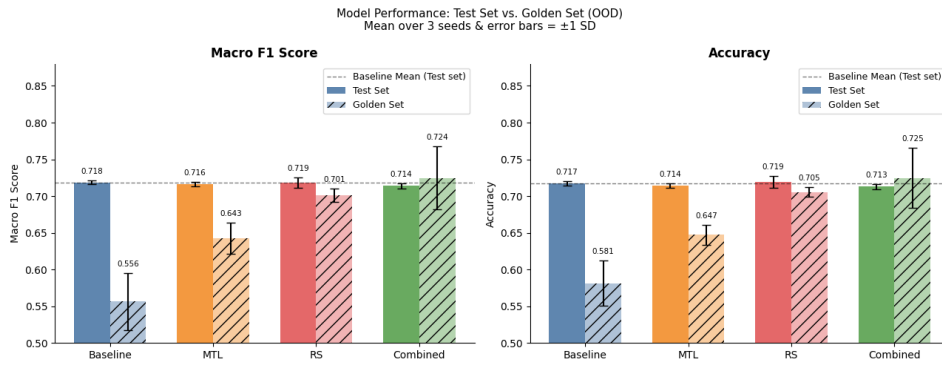


Figure 6: Performance Comparison Chart: Mean Macro-F1 (left) and accuracy (right) on the test set (solid) and golden set (hatched), ± 1 SD

The most striking pattern in Table 2 is the contrast between the two evaluation sets. On the **in-distribution test set**, all four models perform

very similarly, with Macro-F1 scores ranging only from 0.714 to 0.719. McNemar’s test (Appendix 8) further shows that MTL and Combined introduce small but statistically significant regressions relative to the Baseline, while RS remains statistically comparable.

Meanwhile, the **golden set** reveals a substantially different pattern. The baseline drops from $F1 = 0.719$ to 0.557, a 23% relative decline that highlights the model’s difficulty with sarcastic text. Each auxiliary model progressively closes this gap. MTL reaches $F1 = 0.643$ (+0.086 over baseline), RS reaches 0.701 (+0.144), and Combined achieves the highest at 0.724 (+0.168), a 30% relative improvement over the baseline on OOD data.

A POSSIBLE TEST–GOLD TRADE-OFF WITHIN MODELS. Per-seed results (Table 11, Appendix) suggest a possible trade-off between in-distribution benchmark performance and sarcasm-focused robustness within the auxiliary models. In several cases, seeds with higher Golden Set F1 achieve slightly lower test F1, particularly in the Combined model. Although this pattern is only exploratory given the small number of seeds, it suggests that optimizing for standard benchmark performance may not always align with robustness under sarcasm-heavy distribution shift.

5.2 Reliability and Statistical Significance

To account for the small size of the Golden Set ($n = 86$), we employ McNemar’s test (Table 3) to evaluate whether prediction differences between models are statistically reliable. In this challenging OOD setting, RS and Combined both demonstrate significant improvements over the baseline ($p = 0.008$ and $p = 0.004$ respectively). MTL, by contrast, does not reach significance ($p = 0.221$), confirming that its gains are too inconsistent to be statistically reliable on this evaluation set. Equally important, the Combined vs. RS comparison is not significant ($p = 1.000$), indicating that despite Combined’s numerically higher mean F1, the two models are statistically indistinguishable on the Golden Set.

Table 3: McNemar’s test on the Golden Set (median p -value across three seeds).

Model A	Model B	Rescues	Regressions	Net	p	Sig.
MTL	Baseline	5	1	+4	0.221	×
RS	Baseline	9	0	+9	0.008	✓
Combined	Baseline	10	0	+10	0.004	✓
Combined	MTL	7	0	+7	0.023	✓
Combined	RS	2	1	+1	1.000	×

Note: Rescues = Model A correct and Model B wrong; Regressions = Model A wrong and Model B correct.

A 10,000-iteration bootstrap on 3-seed ensemble predictions (Table 9, Appendix) provides a complementary reliability check. After Holm–Bonferroni correction, all auxiliary models outperform the Baseline, although the evidence for MTL remains weaker than for RS and Combined. Combined achieves the highest ensemble Macro-F1 (0.722, 95% CI [0.610, 0.815]), followed by RS (0.694, [0.581, 0.790]) and MTL (0.652, [0.541, 0.750]), while the Baseline remains lowest (0.569, [0.447, 0.676]). Figure 7 visualizes these bootstrap distributions. Despite the wide confidence intervals caused by the small Golden Set ($n = 86$), Combined and RS show a consistent rightward shift, suggesting that their gains are not driven by only a few fortunate predictions. MTL, however, still overlaps substantially with the Baseline.

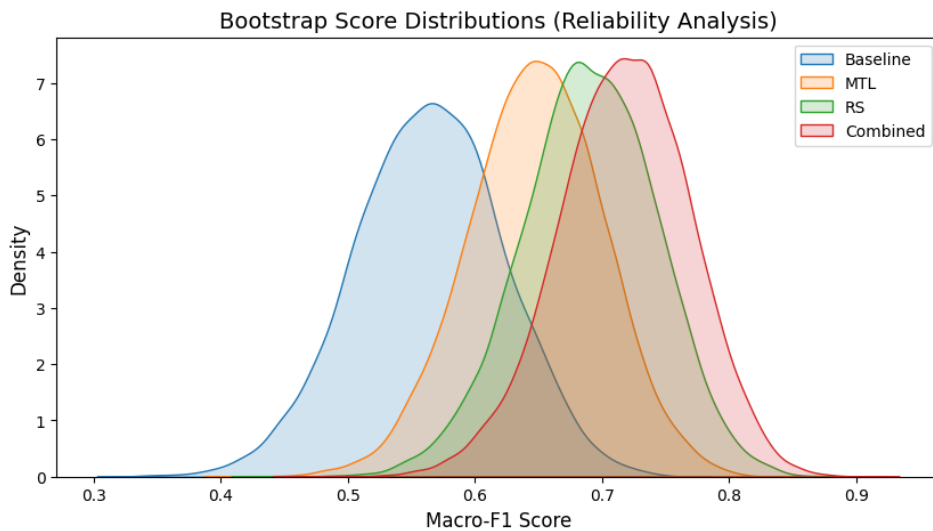


Figure 7: Bootstrap Macro-F1 distributions (10,000 iterations) on the Golden Set for 3-seed ensemble predictions.

Taken together, the statistical evidence confirms that RS and Combined provide robust generalization beyond the baseline, while MTL’s gains remain modest and statistically unstable given the evaluation set size.

5.3 Error Analysis

5.3.1 Per-Class and Polarity Flip Analysis

Aggregate Macro-F1 scores conceal class-level disparities that are critical for understanding how each model handles sarcasm. Figure 8 decomposes performance into per-class F1 on both sets (see Appendix F for confusion matrices).

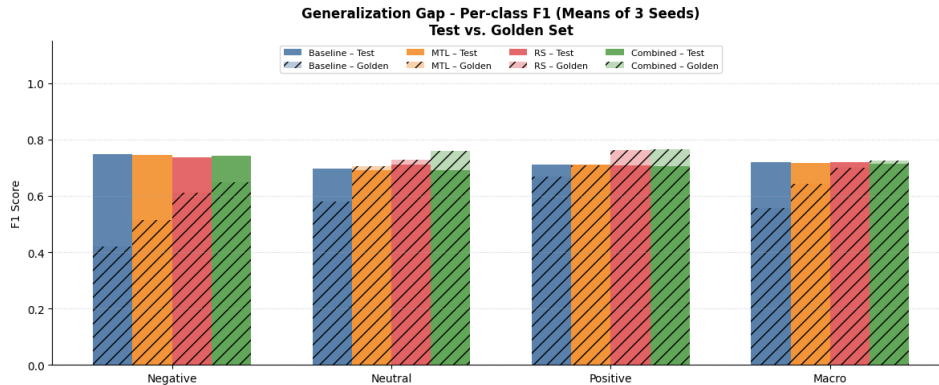


Figure 8: Per-class and Macro-F1 for all models on the test set (solid) and golden set (hatched), averaged across three seeds.

On the test set, all models perform comparably across all three classes, with Neutral being the hardest ($F_1 \approx 0.69$ – 0.71) consistent with its weaker surface-level lexical signals. The Golden Set, however, reveals a fundamentally different failure pattern. The Baseline performance on the Negative class collapses from $F_1 \approx 0.75$ (Test) to $F_1 \approx 0.42$ (Gold). This reflects a surface-level lexical failure: literal positive words in sarcastic tweets are taken at face value, leading to misclassification. The auxiliary models progressively restore this performance: MTL increases Negative F_1 to 0.515, RS to 0.613, and Combined to 0.649. Notably, this restoration of the negative class occurs without degrading performance on the positive class, which maintains a stable $F_1 \approx 0.67$ – 0.76 across all models.

Table 4 quantifies error patterns using Cohen’s h to measure reductions in polarity flips, the most severe error type where Negative is predicted as Positive or vice versa. The Baseline flip rate of 27.9% is only marginally improved by MTL ($h = -0.062$). In contrast, RS and Combined reduce this to around 19% ($h \approx -0.20$), indicating a small-to-medium reduction. The near-identical effect sizes for RS and Combined align with the McNemar results, suggesting that adding sarcasm supervision on top of rationale supervision provides limited additional benefit.

Table 4: Error taxonomy on the Golden Set (aggregated across three seeds).

Model	Error Rate	Polarity Flip	Under-sensitive	Over-sensitive	Cohen’s h
Baseline	41.9%	27.9%	8.5%	5.4%	—
MTL	35.3%	25.2%	7.0%	3.1%	−0.062
RS	29.5%	19.8%	7.4%	2.3%	−0.192
Combined	27.5%	19.4%	5.8%	2.3%	−0.202

Note: Polarity flips = Negative $\leftrightarrow$ Positive; Under-sensitive = Positive/Negative predicted as Neutral; Over-sensitive = Neutral predicted as Positive/Negative. Cohen’s h : $|h| < 0.2$ = small, 0.2–0.5 = medium, > 0.5 = large.

5.3.2 Confidence on Polarity Flip Errors

A well-generalizing model should not only make fewer errors but also be less certain in wrong predictions. Figure 9 shows model confidence on polarity flip errors specifically.

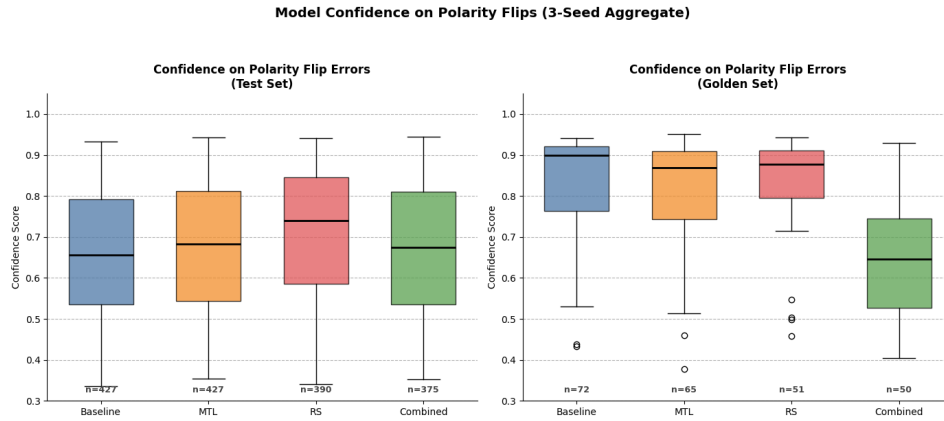


Figure 9: Confidence on polarity flip errors (3-seed aggregate) for the test set (left) and Golden Set (right). Lower confidence reflects better calibration on polarity-flip errors.

On the test set, distributions are similar across all models. On the Golden Set, a clear divergence emerges: the Baseline produces median confidence of ≈ 0.90 on polarity flips, highly confident despite being incorrect. Combined’s median drops to ≈ 0.65 with a wider IQR, meaning it is not only making fewer of these errors but is also more uncertain when it does. This is consistent with Combined’s ECE of 0.071 vs. the Baseline’s 0.226 on Golden Set (full calibration analysis in Appendix G).

5.3.3 Qualitative Analysis via Integrated Gradients

To understand the internal mechanisms behind model behavior, Integrated Gradients (IG) attribution was applied to three representative golden-set cases. Figures for all cases are in the Appendix H; key findings are summarized here.

Case 1: All auxiliary models succeed; Baseline fails. The tweet “*Oh no I have to do a 10-slide presentation on why I love Brixton [...] and I got NOTHING*” is Negative. The Baseline assigns high attribution to positive cues such as “*love*” and the emoticon “*:-)*”. In contrast, all auxiliary models assign stronger attribution to “*Oh no*” and “*NOTHING*”. This suggests that auxiliary supervision helps the model move beyond isolated positive words and incorporate the broader contextual cues that reveal the speaker’s true sentiment.

Case 2: RS and Combined succeed; Baseline and MTL fail. In the tweet “*Who needs sleep? It’s not like I have a test...*”, the Baseline and MTL models assign high attribution to the rhetorical structure and predict Neutral. RS and the Combined model instead attend to cues such as “*test*”, “*tomorrow*”, and “*not*”, leading to the correct prediction. This suggests that rationale-based supervision helps the model attend to contextual cues relevant for pragmatic interpretation, while the Combined model shows a similar shift away from the misleading rhetorical structure.

Case 3: All models fail. The tweet “*@MetroNorth wall to wall people on the platform [...] Thanks for the Sat. Sched. Great sense*” is Negative. The Baseline, MTL, and RS models all predict Positive, focusing on lexical cues such as “*Thanks*”, “*Great*”, and “*sense*”. The Combined model at least did not flip the polarity by predicting Neutral, but still assigns substantial attribution to “*Great sense*”, indicating uncertainty rather than true understanding. Correctly interpreting this tweet requires world knowledge that a Saturday schedule implies reduced train service and overcrowding on a workday. This limiting case suggests that while auxiliary supervision can reduce surface-level errors, it remains insufficient when sarcasm depends on external knowledge not present in the text.

5.3.4 Diagnostic and Representation Analyses

These analyses reveal a clear distinction between task-level performance and representational change.

SARCASM HEAD DIAGNOSTIC. On the fully sarcastic Golden Set, the sarcasm heads of the MTL and Combined models detect sarcasm in only 41.9% and 47.7% of cases, respectively (Table 5). Surprisingly, sentiment accuracy is *lower* when sarcasm is correctly detected than when it is missed

(Combined: 0.585 vs. 0.872, $p = 0.009$). This suggests that the sarcasm head does not act as a reliable corrective signal. It may identify more difficult sarcastic cases rather than consistently supporting correct sentiment classification, which helps explain the modest gains of MTL.

Table 5: Sarcasm head diagnostics on the Golden Set ($n = 86$).

Model	Sarcasm Detection Rate	Sentiment Acc. (Sarcasm Detected)	Sentiment Acc. (Sarcasm Missed)	p -value
MTL	41.9%	0.537	0.720	0.0743
Combined	47.7%	0.585	0.872	0.0089**

Note: Sentiment accuracy is lower when sarcasm is correctly detected, suggesting internal signal conflict. ** $p < 0.01$.

REPRESENTATIONAL RESTRUCTURING: CKA AND T-SNE. Despite the inconsistent performance of the sarcasm head, CKA and t-SNE results indicate that rationale-based supervision changes the encoder representation more substantially than MTL (Table 6). The high CKA for MTL (0.90) indicates that its feature space remains geometrically similar to the Baseline, which is consistent with its modest OOD gains. Conversely, the Combined model’s lower CKA (0.71) indicates stronger representational restructuring.

Table 6: Centered Kernel Alignment (CKA) of each model’s encoder representations relative to the Baseline on the Golden Set (mean $\pm$ SD across seeds). Lower CKA indicates greater representational divergence from the baseline.

Model	Mean CKA	Interpretation
MTL	0.9038 ± 0.024	Low Divergence
RS	0.8282 ± 0.028	Moderate Restructuring
Combined	0.7080 ± 0.077	Strong Restructuring

This pattern is also reflected in the t-SNE geometry (Figure 10). The Combined model achieves the clearest separation (12.9), with sarcastic-negative samples more clearly separated from genuine-positive ones. Interestingly, although MTL achieves a higher numerical separation score (8.7) than RS (7.7), the visualization still shows fragmented and overlapping clusters. In contrast, the RS and Combined models exhibit a more organized geometry with clearer separation between the two groups. Together, these results suggest that the strongest improvements are more closely associated with rationale supervision than with the sarcasm head alone.



Figure 10: t-SNE visualization of encoder representations on the golden set. Red triangles = sarcastic-negative samples (OOD); green circles = genuinely positive samples (in-distribution). Separation scores (higher = better) are shown in each panel title.

6 DISCUSSION

6.1 Summary of Key Findings: Answer to the Main Research Question

The goal of this thesis is to investigate whether auxiliary supervision can improve the robustness of a Twitter-adapted RoBERTa sentiment model under sarcasm-heavy distribution shift. Therefore, our main question asked: **To what extent can auxiliary supervision through sarcasm detection, rationale supervision, and their combination improve the out-of-distribution robustness of a RoBERTa-based sentiment model on sarcastic tweets?**

The results show that auxiliary supervision can partially reduce sarcasm-related robustness failures, but not all supervision signals are equally effective. RS and the Combined model substantially improve sarcasm-heavy OOD performance, whereas MTL provides smaller and less reliable gains. At the same time, in-distribution TweetEval performance remains almost unchanged across models. This asymmetry is expected because the auxil-

iary objectives specifically target sarcasm failures caused by surface-literal bias, which appear mainly on sarcasm-heavy OOD data.

6.2 RQ1: Baseline Failure on Sarcastic Tweets

RQ1 examined how the baseline RoBERTa model performs on standard and sarcasm-heavy data, and what systematic errors it makes on sarcastic tweets. The baseline achieves a competitive TweetEval F1 of 0.719, consistent with He (2024) on the same dataset and backbone, but drops to 0.557 on the Golden Set, a 23% relative decline. The failure is concentrated in sarcastic-negative tweets: Negative-class F1 falls from 0.75 to 0.42, and 27.9% of Golden Set predictions become polarity flips.

The errors reflect a systematic lexical-bias failure. Figure 10 is consistent with this failure: the baseline has the weakest separation between sarcastic-negative and genuine-positive tweets among all models. The qualitative examples show the same pattern: the model overweights positive cues such as “Great” and “love” despite negative intent. This happens because such phrases are overwhelmingly positive in standard training data, while sarcasm makes these surface cues unreliable. The pattern is consistent with prior work showing that sarcasm requires pragmatic interpretation beyond literal wording (Joshi et al., 2017). Here, the problem is substantial rather than occasional: more than one in four sarcastic samples results in polarity reversal, indicating that sarcasm is a major robustness failure mode rather than a set of isolated errors.

6.3 RQ2: MTL with Sarcasm Detection Provides Limited Gains

RQ2 examined whether MTL with sarcasm detection improves sentiment classification on sarcasm-heavy data. The improvement is only modest. MTL increases Golden Set F1 by 0.086, but the gain is not statistically significant ($p = 0.221$), and polarity-flip errors are only slightly reduced.

The diagnostic analysis helps explain why. Surprisingly, sentiment accuracy is lower when sarcasm is correctly detected (0.537) than when it is missed (0.720). Sarcasm detection may flag that literal sentiment is unreliable, but it does not consistently help the model recover the intended sentiment. As a result, the auxiliary sarcasm signal may introduce task conflict rather than reliable correction, suggesting a form of negative transfer, a common challenge in multi-task learning (Crawshaw, 2020).

This finding contrasts with prior work by Majumder et al. (2019) and Tan et al. (2023), who reported more consistent gains from sarcasm-aware MTL. A likely explanation is encoder strength. Earlier studies relied mainly on GRU and Bi-LSTM architectures, while RoBERTa already captures

richer contextual dependencies through self-attention. This may reduce the marginal benefit of sarcasm detection when the base encoder is already strong. CKA further supports this interpretation: MTL remains highly similar to the baseline representation space (0.90), indicating only limited representational change despite the additional task signal.

6.4 RQ3: Rationale Supervision (RS) Provides Reliable Robustness Gains

RQ3 examined how rationale supervision affects performance on sarcastic tweets. RS produces the most reliable robustness improvement, reaching 0.701 Golden Set F1 with the lowest seed variance ($SD = 0.009$) and significantly outperforming the baseline ($p = 0.008$). For reference, the best model on the same sarcasm benchmark reported 58.16 under the official positive-negative F1 metric (Rosenthal et al., 2014), while RS reaches 68.8 under the same metric. Although this is not a direct comparison because the models and training setups differ substantially, it suggests that the improvement is large relative to a well-known sarcasm benchmark.

The critical question is whether these gains reflect genuine representational change or simply incidental regularization from the auxiliary head. Several findings support the former. If the effect were mainly regularization, improvements would be expected across both evaluation sets. Instead, RS leaves in-distribution performance nearly unchanged while substantially improving Golden Set F1 by +0.144, suggesting that the gains are concentrated on sarcasm-related failures rather than general performance improvements. This targeted improvement is also reflected in the representation analyses: RS shows moderate divergence from the baseline ($CKA = 0.828$), and the t-SNE results reveal clearer separation between sarcastic-negative and genuinely positive samples, which is the boundary where the baseline most consistently fails. Together, these findings suggest that RS changes sentiment representations rather than merely smoothing predictions. This is consistent with Wiegreffe et al. (2021), who argue that rationale supervision is most useful when rationales provide information beyond the label itself, which our prompt constraints were specifically designed to ensure.

However, RS is not without limits. The qualitative analysis (5.3.3 Case 3) shows that when sarcasm depends on external world knowledge absent from the text, such as knowing that a Saturday train schedule implies reduced service, rationale supervision cannot fully compensate. This suggests that RS is most effective when pragmatic meaning can be inferred from the text itself, but remains limited when interpretation depends on external knowledge.

6.5 RQ4: Combination Improves Peak OOD Performance but Adds Instability

RQ4 examined whether combining sarcasm detection and rationale supervision improves performance beyond either approach alone. The results show that the Combined model improves peak robustness, but introduces instability. The Combined model achieves the highest Golden Set F1 (0.724), the lowest Golden Set ECE (0.071), and the clearest t-SNE separation between sarcastic-negative and genuinely positive tweets. However, its improvement over RS is not statistically significant ($p = 1.000$), and its seed variance is much higher (SD = 0.043 vs. 0.009). More importantly, it performs significantly worse than both the Baseline and RS on the in-distribution test set ($p = 0.0295$ and $p = 0.0007$).

Diagnostic analyses reveal persistent interference from the sarcasm signal. In the Combined model, sentiment accuracy is lower when sarcasm is correctly detected than when it is missed (0.585 vs. 0.872, $p = 0.009$), suggesting that sarcasm features can still conflict with sentiment prediction. At the same time, the Combined model shows the strongest representational shift from the Baseline, with the lowest CKA score (0.708) and the clearest separation between sarcastic-negative and genuinely positive samples. The most careful interpretation is therefore not clear synergy, but a stronger yet less stable representation space. Practically, RS appears to be the safer standalone choice when stability matters, while Combined may be considered only when maximizing peak OOD robustness is prioritized over stability.

6.6 RQ5: Statistical Reliability

Because the Golden Set is relatively small ($n = 86$), per-seed F1 fluctuations are large enough to affect model ranking. Bootstrap resampling, McNemar’s test, and Holm–Bonferroni correction together show that RS and Combined provide statistically reliable improvements, while MTL’s gains remain inconsistent at the sample level despite modest average improvements.

6.7 Scientific and Societal Implications

Scientifically, this study makes three main contributions. First, to the best of our knowledge, it provides the first controlled multi-seed comparison of sarcasm-task MTL, rationale supervision, and their combination within the same RoBERTa framework on sarcasm-heavy OOD data. Second, it identifies a potential boundary condition for MTL: sarcasm detection may provide diminishing returns when the base encoder already captures rich

contextual information, helping explain why gains reported in earlier GRU- and Bi-LSTM-based studies are weaker in this RoBERTa-based setting. Third, it shows that rationale supervision reshapes the model’s representation space more substantially than task-based supervision. More broadly, the findings suggest that robustness to pragmatic language may require supervision over reasoning patterns rather than only additional labels.

These findings also have broader societal implications because sentiment analysis increasingly supports automated decision-making in areas such as political opinion tracking and customer feedback monitoring. When sarcastic criticism is misinterpreted as positive sentiment, public dissatisfaction may be underestimated and important negative feedback overlooked. At scale, this can distort automated interpretations of public opinion. The RS model reduces these failures while preserving standard benchmark performance, and does so without requiring sarcasm labels or LLM access during inference, making it the most practical robustness strategy evaluated in this thesis.

6.8 Limitations and Future Directions

Several limitations should be acknowledged. First, the Golden Set is relatively small ($n = 86$), limiting statistical power and making it difficult to distinguish reliably between RS and Combined. In addition, although the Golden Set is never used for training or checkpoint selection, it is used for auxiliary-weight selection. Because the Golden Set is already small, creating separate development and test splits was considered impractical. As a result, the reported robustness estimates may be slightly optimistic. However, this mainly affects absolute performance estimates rather than the overall pattern of findings. As shown in Appendix E.1, RS remains consistently stable across λ values, while MTL remains the weakest auxiliary model throughout the evaluated range. This suggests that the main conclusion – that rationale supervision is more effective than sarcasm-task MTL for sarcasm robustness – is not driven solely by λ selection. Future work would therefore benefit from larger sarcasm-focused benchmarks with separate development and test splits for cleaner robustness evaluation under distribution shift.

Second, the effectiveness of rationale supervision is limited by both rationale quality and scale. The rationales were generated automatically using GPT-4o-mini without human validation, and qualitative failure cases suggest that rationale supervision remains limited when sarcasm depends on external world knowledge absent from the text. In addition, rationales were generated only for the supervised training data, which is small compared to the 124 million tweets used during backbone pretraining. Future

work could therefore explore larger-scale rationale generation, stronger reasoning models, or human-written rationales to improve pragmatic reasoning more consistently.

Third, the study evaluates auxiliary supervision using only a RoBERTa-base backbone. The weak contribution of MTL in this setting suggests that the effectiveness of auxiliary supervision may depend on encoder strength and model scale. Future work should therefore examine whether the same supervision strategies behave differently in weaker architectures, larger language models, or inference-time reasoning setups such as chain-of-thought prompting (Wei et al., 2022).

7 CONCLUSION

This thesis started from the observation that even a strong RoBERTa-based sentiment model can struggle on sarcastic tweets. Error analyses showed that sarcastic tweets often contain positive wording despite expressing negative sentiment, making patterns learned from standard sentiment data unreliable under sarcasm. This led to the investigation of whether auxiliary supervision can improve robustness.

The results show that auxiliary supervision can reduce these failures, but the type of signal matters. MTL with sarcasm detection provides modest but statistically unreliable gains. The Combined model achieves the highest Golden Set performance, but its improvement over RS is not statistically significant and comes with higher seed variance.

Rationale Supervision therefore emerges as the strongest overall approach. RS improves Golden Set F1 to 0.701, significantly outperforms the baseline, and substantially reduces polarity-flip errors. Its representational divergence from the baseline suggests that RS does more than regularize the model: it helps the encoder capture pragmatic sentiment cues that are missing from label-only training.

Overall, this thesis contributes to understanding sarcasm robustness as a supervision problem. For pragmatic language, teaching a model why the literal reading is misleading appears more effective than only teaching it that sarcasm is present. RS is also practically attractive because rationales are only required during training rather than inference, making the approach feasible for real-world social media systems.

Future work should evaluate these findings on larger sarcasm-focused benchmarks, validate rationale quality through human annotation, and examine whether robustness gains generalize across different social groups, dialects, and communication contexts.

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A SOFTWARE AND PACKAGES

Table 7 lists the main software packages used in the experimental environment.

Table 7: Software and packages used in the experimental environment.

Package Category	Version
Python	3.x
NumPy	2.2.6
Pandas	2.3.3
Scikit-learn	1.7.2
Matplotlib	3.10.8
Seaborn	0.13.2
PyTorch	2.5.1+cu121
Transformers	5.3.0
SentenceTransformers	5.3.0
Hugging Face Hub	1.6.0
Optuna	4.7.0
OpenAI API	2.30.0
emoji	2.15.0
TextBlob	0.19.0
tqdm	4.67.3

B PROMPT TEMPLATES

The following prompts were used to generate rationale annotations via the OpenAI Batch API (GPT-4o-mini, temperature = 0.1).

Sentiment Rationale Prompt

You are an expert sentiment annotator for social media. You will be provided with a Tweet and its human-verified sentiment label (Positive, Negative, Neutral). Your task is to generate a single, logical sentence explaining WHY the tweet matches this ground truth sentiment based on linguistic cues.

Strict rules:

1. Lexical Grounding: You MUST quote specific words from the tweet.
2. Sarcasm Handling: If the sentiment relies on sarcasm, explicitly state the contrast between the literal words and the intended meaning.
3. Direct Start: Begin directly with the linguistic evidence.
4. Length: under 20 words.
5. Anti-Leakage: Do NOT use the label word unless necessary.

Return only valid JSON: {"rationale": "<Your justification>"}

Sarcasm Rationale Prompt

You are an expert sarcasm annotator for social media text. You will be provided with a Tweet and its ground truth sarcasm label (Sarcastic or Non-Sarcastic). Your task is to generate a single, logical sentence explaining the linguistic mechanics of the tweet.

Strict rules:

1. If Sarcastic: Explicitly explain the semantic contrast between literal meaning and contextual reality.
2. If Non-Sarcastic: Explain how the literal words directly convey the intended meaning without hidden contrast.
3. No Fluff: Start directly with the linguistic breakdown.
4. Length: under 20 words.
5. Anti-Leakage: Do NOT repeat the label word itself unless necessary.

Return only valid JSON: {"rationale": "<Your justification>"}

C EXPLORATORY DATA ANALYSIS (EDA) RESULT

This appendix reports additional EDA results used to support preprocessing and modeling decisions.

c.1 *Dataset Distribution and Length*

Figure 11 compares the label distribution of the standard TweetEval test set and the sarcasm-heavy Golden Set. The Golden Set contains substantially more negative tweets (46.5% vs. 32.3%), supporting its use as a robustness-focused sarcasm benchmark rather than a standard sentiment test set.

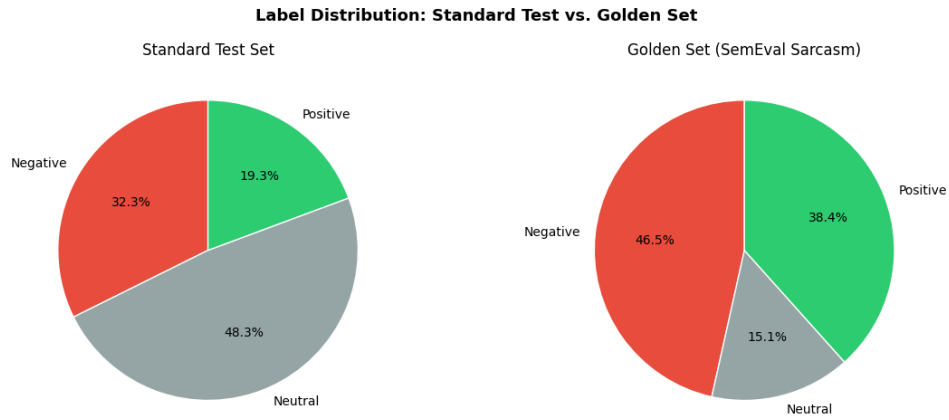


Figure 11: Label distribution of the standard TweetEval test set and the Golden Set.

c.2 *Rationale Quality Checks*

Figures 12 and 13 show the rationale length distributions for the sentiment and sarcasm datasets. Most rationales remain within the intended prompt constraint range.

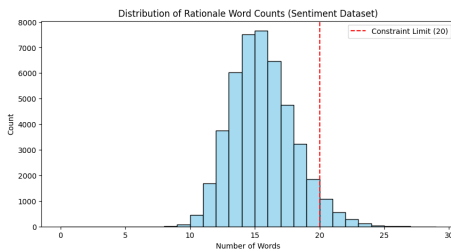


Figure 12: Length distribution of generated sentiment rationales.

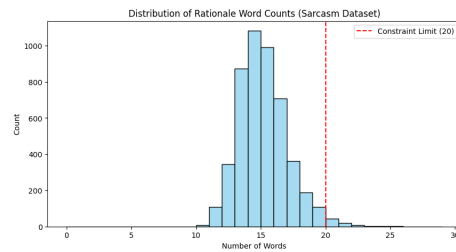


Figure 13: Length distribution of generated sarcasm rationales.

Semantic similarity analysis using all-MiniLM-L6-v2 produced moderate cosine similarity scores (0.45 for sentiment; 0.49 for sarcasm), suggesting that rationales remain grounded in the source tweets while contributing additional semantic information.

D MCNEMAR’S TEST RESULTS ON THE TEST SET

Table 8: McNemar’s test results on the test set (median p -value across three seeds).

Model A	vs. Model B	Rescues	Regressions	Net	p -value
MTL	Baseline	363	423	−60	0.0294*
RS	Baseline	449	404	+45	0.1198
Combined	Baseline	435	479	−44	0.0295*
Combined	MTL	389	393	−4	0.4177
Combined	RS	459	574	−115	0.0007***

*Note: Rescues = A correct & B wrong; Regressions = A wrong & B correct. * $p < 0.05$, ***

*$p < 0.01$, *** $p < 0.001$.*

Table 9: Bootstrap on the Sarcasm-Heavy Golden Set. Results include Ensemble Macro-F1, Bootstrap 95% Confidence Intervals (CI), and p -values adjusted via the Holm-Bonferroni method.

Model	Ensemble F1	95% CI	CI Width	p -value	Sig. (Holm)
Combined	0.7215	[0.610, 0.815]	0.2045	0.0001***	Yes
RS	0.6943	[0.581, 0.790]	0.2082	0.0013**	Yes
MTL	0.6521	[0.541, 0.750]	0.2084	0.0122*	Yes
Baseline	0.5690	[0.447, 0.676]	0.2290	1.0000	No

*Note: p -values indicate significance relative to the Baseline. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Confidence intervals were estimated via bootstrap resampling ($R = 10,000$).*

E PER-LAMBDA AND PER-SEED RESULTS

E.1 *Per-Lambda Result*

Table 10: Lambda selection results for MTL, RS, and Combined models. Score = $0.4 \times \text{Val F1} + 0.6 \times \text{Gold F1}$. Values reported as mean $\pm$ SD across three seeds. Bold rows indicate the selected configuration.

Model	λ	Val F1	Gold F1	Score
MTL	0.10	0.759 $\pm$ 0.003	0.582 $\pm$ 0.044	0.652 $\pm$ 0.027
MTL	0.15	0.758 $\pm$ 0.005	0.577 $\pm$ 0.045	0.649 $\pm$ 0.027
MTL	0.20	0.757 $\pm$ 0.001	0.645 $\pm$ 0.008	0.690 $\pm$ 0.005
MTL	0.25	0.757 $\pm$ 0.003	0.639 $\pm$ 0.003	0.686 $\pm$ 0.003
MTL	0.30	0.758 $\pm$ 0.005	0.617 $\pm$ 0.006	0.673 $\pm$ 0.003
RS	0.10	0.759 $\pm$ 0.001	0.701 $\pm$ 0.009	0.724 $\pm$ 0.005
RS	0.15	0.756 $\pm$ 0.002	0.694 $\pm$ 0.015	0.719 $\pm$ 0.008
RS	0.20	0.758 $\pm$ 0.001	0.695 $\pm$ 0.014	0.720 $\pm$ 0.008
RS	0.25	0.759 $\pm$ 0.002	0.695 $\pm$ 0.014	0.721 $\pm$ 0.008
RS	0.30	0.758 $\pm$ 0.002	0.696 $\pm$ 0.017	0.721 $\pm$ 0.009
Combined	0.10	0.756 $\pm$ 0.002	0.712 $\pm$ 0.031	0.730 $\pm$ 0.019
Combined	0.15	0.757 $\pm$ 0.002	0.721 $\pm$ 0.039	0.735 $\pm$ 0.023
Combined	0.20	0.755 $\pm$ 0.001	0.724 $\pm$ 0.043	0.737 $\pm$ 0.026
Combined	0.25	0.756 $\pm$ 0.003	0.706 $\pm$ 0.057	0.726 $\pm$ 0.034
Combined	0.30	0.756 $\pm$ 0.002	0.711 $\pm$ 0.064	0.729 $\pm$ 0.038

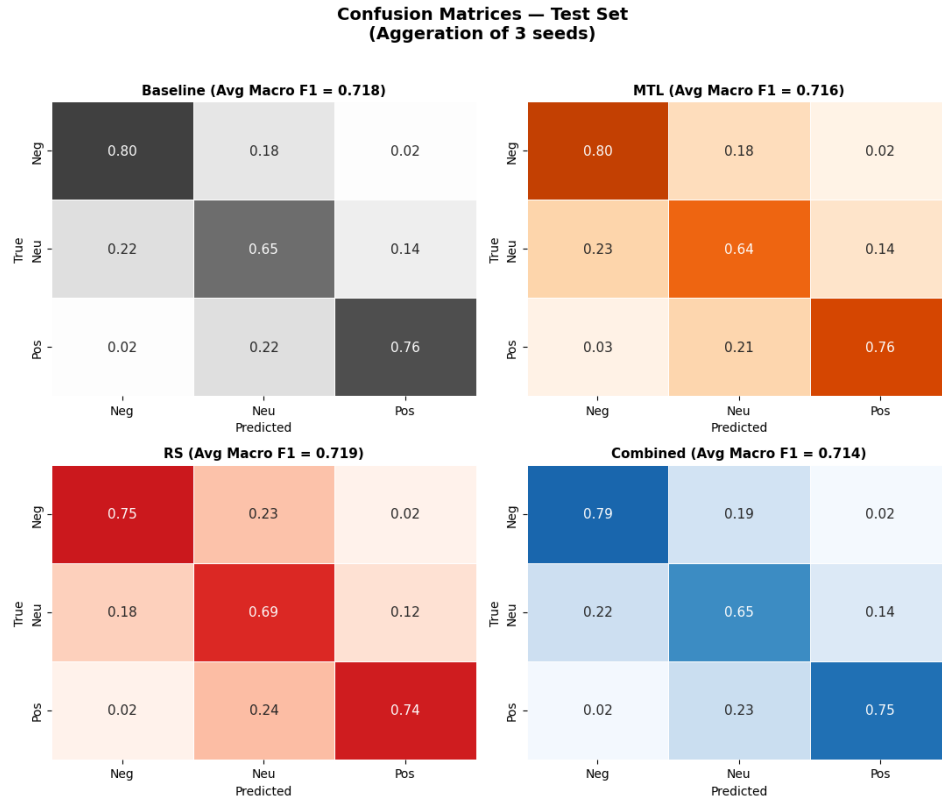
E.2 *Per-Seed Result*

Table 11 reports per-seed Macro-F1 for all models on both evaluation sets. A consistent pattern is visible: within the same model, seeds that achieve higher golden-set F1 tend to produce slightly lower test-set F1, and vice versa.

Table 11: Per-seed Test and Golden Set performance for all models

Model	Seed	Test F1 $\uparrow$	Gold F1 $\uparrow$
Baseline	100	0.7211	0.5878
	42	0.7162	0.5690
	7	0.7181	0.5126
MTL	7	0.7125	0.6654
	100	0.7164	0.6402
	42	0.7185	0.6230
RS	100	0.7260	0.6925
	7	0.7174	0.7108
	42	0.7123	0.6998
Combined	42	0.7158	0.6830
	100	0.7155	0.7215
	7	0.7099	0.7687

F CONFUSION MATRICES

Figure 14: Confusion matrices on the **test set** (3-seed aggregate, row-normalized).

Confusion Matrices — Golden Set (OOD)
(Aggeration of 3 seeds)

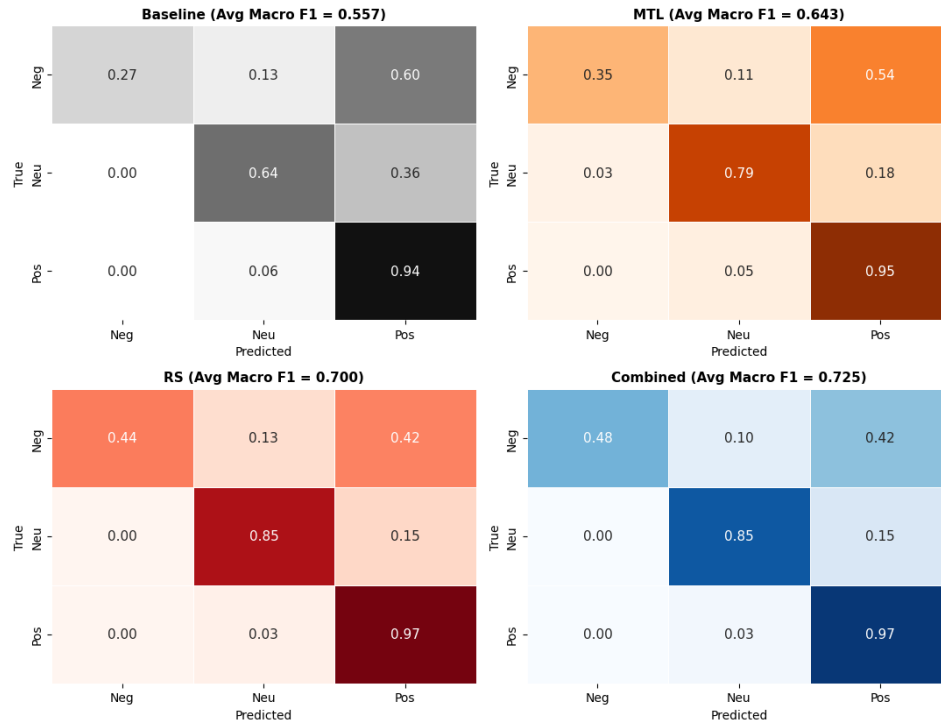


Figure 15: Confusion matrices on the **Golden Set**

G MODEL CALIBRATION/RELIABILITY DIAGRAM

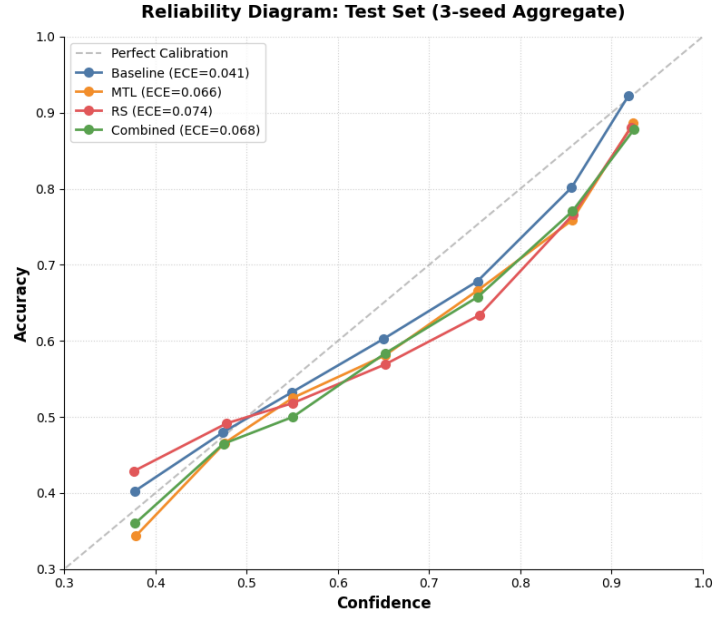


Figure 16: Reliability diagram on the test set (3-seed aggregate). The dashed diagonal indicates perfect calibration. ECE values are reported in the legend; lower is better.

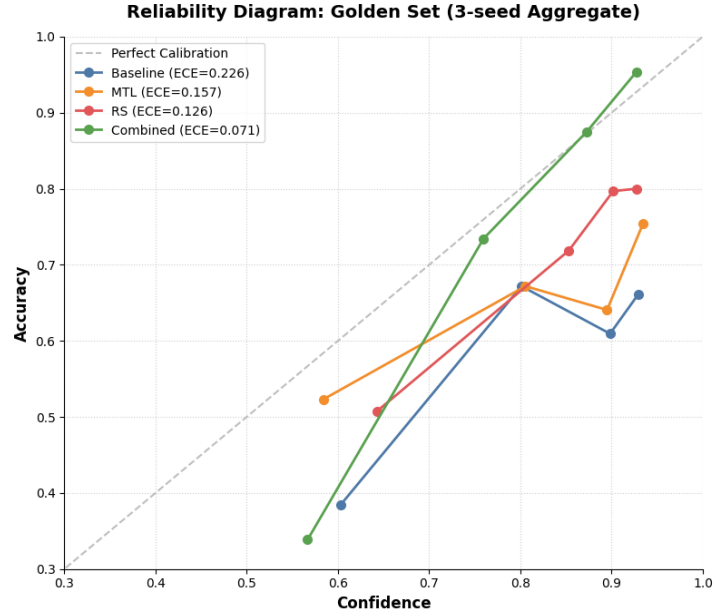


Figure 17: Reliability diagram on the Golden Set (3-seed aggregate)

H INTEGRATED GRADIENTS ATTRIBUTION PLOTS

Figures 18–20 show token-level Integrated Gradients attributions for the three qualitative cases. Coloured bars indicate tokens with normalised attribution above 0.30; grey bars are low-attribution tokens. Attributions are normalised to the maximum score within each model and case.

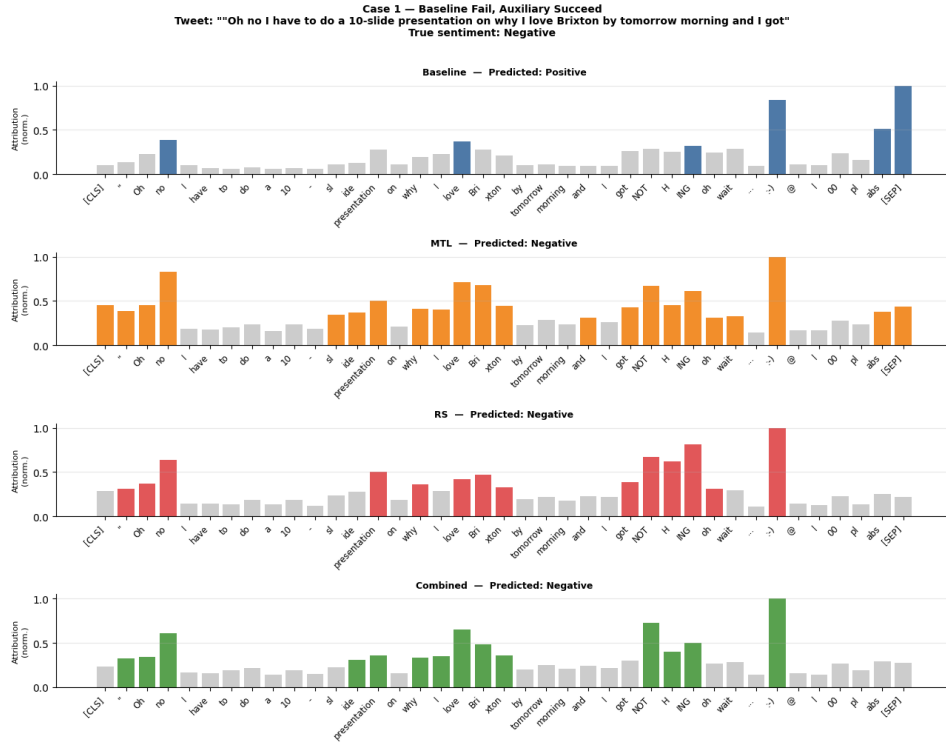


Figure 18: **Case 1 — All auxiliary models succeed; Baseline fails.** Tweet: "Oh no I have to do a 10-slide presentation on why I love Brixton [...] and I got NOTHING"

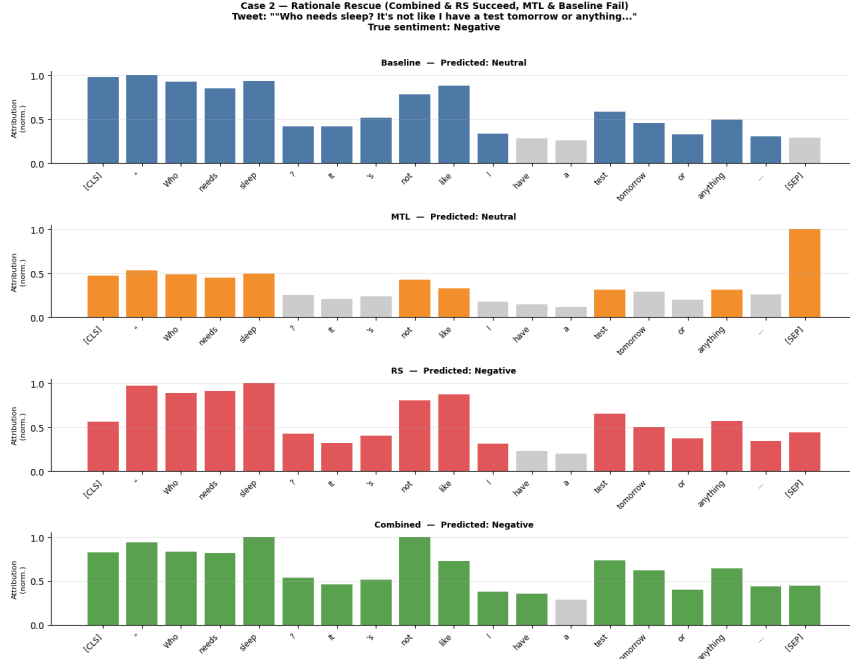


Figure 19: **Case 2 — Rationale rescue; MTL and Baseline fail.** Tweet: "Who needs sleep? It's not like I have a test..."

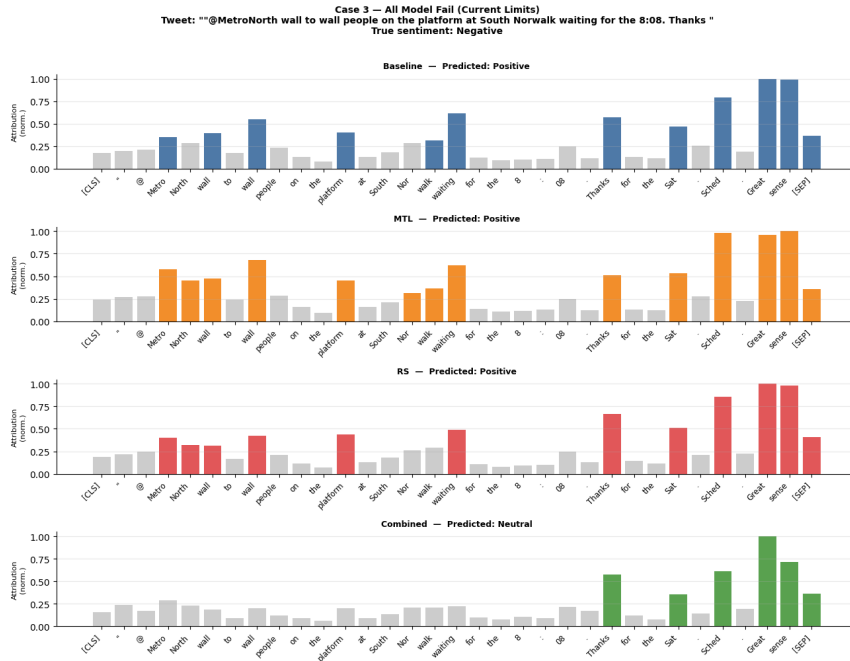


Figure 20: **Case 3 — All models fail; Combined partially recovers.** Tweet: "@MetroNorth wall to wall people on the platform [...] Thanks for the Sat. Sched. Great sense"

I EXPLORATORY PILOT EXPERIMENTS

Before the final full-scale training, we conducted exploratory pilot experiments to test whether targeted architectural or optimization changes could improve performance beyond the tuned baselines. Each experiment used a 20% stratified subset of the training data, 1 seed, and 2 training epochs for quicker training.

Table 12: Summary of exploratory pilot experiments on the sarcasm-heavy Golden Set.

Method	Description and Rationale	Score	Decision
Multi-Task Learning (MTL) Variants			
MTL-0: 1:1 cycling	Reference MTL setup alternating 1:1 sentiment and sarcasm batches	0.6108	Reference
MTL-1: SemEval-2018 Irony Augmentation	Adds SemEval-2018 irony data to provide more diverse sarcasm examples.	0.6185 (+0.0077)	Not adopted*
MTL-2: iSarcasm Oversampling	Oversamples iSarcasm by $\times 3$ to reduce the imbalance between the two training tasks.	0.5975 (-0.0133)	Not adopted
MTL-3: Proportional Batch Sampling	Samples sarcasm batches less frequently to reduce over-training on the much smaller sarcasm dataset.	0.5960 (-0.0147)	Not adopted
MTL-4: Focal Loss for Sarcasm Head	Uses focal loss so the sarcasm head focuses more on difficult sarcasm examples.	0.6106 (-0.0002)	Not adopted
MTL-5: Uncertainty-Weighted Multi-Task Loss	Automatically learns the balance between sentiment and sarcasm losses instead of using a fixed λ .	0.6148 (+0.0040)	Not adopted**
Rationale Supervision (RS) Variants			

Method	Description	Score	Decision
RS-0: Parallel Auxiliary Heads	Reference RS setup where sentiment classification and rationale alignment are trained through separate heads.	0.6143	Reference
RS-1: Rationale Concatenation	Concatenates rationale-aligned features with the tweet representation before classification.	0.6236 (+0.0093)	Not adopted***
RS-2: Gated Rationale Fusion	Uses a learnable gate to control how much rationale information contributes to prediction.	0.6145 (+0.0002)	Not adopted**
Combined Model Variants			
Combined-0: Parallel Combined Model	Reference combined setup using both sarcasm supervision and rationale alignment through separate auxiliary heads.	0.6148	Reference
Combined-1: Concatenated Combined Model	Adds rationale-concatenated features to the combined model before prediction.	0.6184 (+0.0036)	Not adopted**

Note: Pilot experiments used a reduced setup (20% subset, 1 seed, 2 epochs) for rapid exploration. The score is computed as $(0.4 \times \text{Test F1}) + (0.6 \times \text{Gold F1})$. Values in parentheses indicate the score change relative to the baseline model of the same variant family.

*Although SemEval augmentation slightly improved the pilot score, it was not adopted due to the risk of negative transfer. SemEval irony labels do not align cleanly with iSarcasm’s binary sarcastic/non-sarcastic formulation, potentially introducing label noise from a mismatched source.

**The improvement was too small to justify additional architectural or optimization complexity.

*** Although rationale concatenation improved pilot performance, the gain did not replicate in the full multi-seed experiments (0.001 lower and higher variance). Therefore, the final RS and Combined models keep the original parallel-head architecture.